

UNIVERSITY OF AMSTERDAM

MASTER'S THESIS

Interventions in Asymmetric Belief Systems

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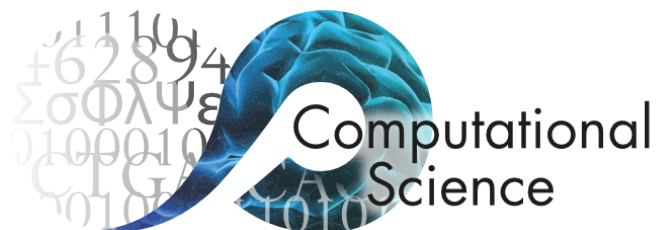
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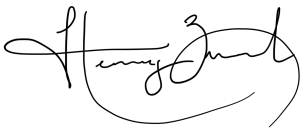


Declaration of Authorship

I, Henry ZWART, declare that this thesis, entitled ‘Interventions in Asymmetric Belief Systems’ and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at the University of Amsterdam.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed:

A handwritten signature in black ink, appearing to read 'Henry Zwart', with a stylized flourish at the end.

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UNIVERSITY OF AMSTERDAM

Abstract

Faculty of Science
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Master of Science in Computational Science

Interventions in Asymmetric Belief Systems

by Henry ZWART

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Thank the people that have helped: supervisors, family, etc.

Use of AI

Statement that generative AI was not used in the writing of this thesis.

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Abbreviations

CSL Computational Science **L**ab

UvA Universiteit van **A**msterdam

Chapter 1

Terminology and notation

Define key terminology, outline notation used in following sections.

Chapter 2

Introduction

2.1 Plan

- Motivate the problem
- Contributions
- Research questions (thinking these are perhaps better left to later, in favour of contributions here)

2.2 Research questions

Theoretical contributions:

- **RF1.1:** Extending the causal attitude network model of belief systems to: (i) support asymmetric causal effects between beliefs and attitudes, and (ii) model intervention dynamics.

Inferring belief systems:

- **RQ2.1:** To what extent are causal relations *symmetric* or *asymmetric* in models of climate change belief systems inferred from observational data?
- **RQ2.2:** How do (symmetric or asymmetric) belief systems relating to climate change vary between conservative and liberal individuals?

Intervention dynamics:

- **RQ3.1:** How do asymmetric and symmetric beliefs systems inferred from the climate attitudes dataset differ with regards to intervention strategy and effectiveness?

- **RQ3.2:** How do intervention outcome and effectiveness vary between individuals with different initial conditions, or between conservative and liberal individuals?

2.3 Contributions

We

1. We
2. We present a mathematical model for belief system dynamics that does not assume equilibrium and does not assume symmetric influence between cognitive aspects (Chapter 3)
3. We describe a novel parameter estimation method for fitting binary models to continuous data (Chapter 4).
4. We calibrate said model to data from a recent longitudinal survey including items on beliefs and attitudes regarding climate change (Chapter 5).
5. We demonstrate, by way of the calibrated model, the existence of asymmetric influence relations between beliefs and attitudes. Furthermore we show that influence relations are not *necessarily* asymmetric, may vary in the degree of asymmetry, and can be unidirectional (Chapter 5).
6. We demonstrate that the decision to represent asymmetric relations in belief system models can change intervention dynamics, and therefore conclusions one draws regarding intervention effect and effectiveness (Chapter 5).
7. We then show that belief systems may vary significantly between individuals, by fitting the proposed model to subsets of the climate attitudes dataset comprising conservative and liberal individuals (Chapter 6).
8. Finally, we show that reasoning about the effects of interventions on *individuals* is, in general, non-trivial. How an individual responds to an intervention typically depends on their prior belief system state, including beliefs and attitudes other than the target and goal of intervention (Chapter 6).

2.4 Proposed thesis structure

- **Terminology and notation**
- **Introduction:**
 - Motivate the problem,
 - Outline contributions (research questions)
- **Asymmetric belief system:**

- Define and illustrate the model
- Model simulation with Glauber dynamics
- How do we model interventions?
- **Methods:**
 - Counterfactual intervention experiments — comparing against the no-intervention scenario, measuring differences in effects.
 - Parameter estimation:
 - Maximum likelihood estimation
 - Conditional on a specific binarisation
 - Marginalising over binarisation process
 - Regularisation
- **Existence and impacts of asymmetry in belief systems**
 - Results for:
 - **RF1:** Show asymmetric model fit
 - **RQ2.1:** Existence of asymmetric relations. Some, but not all, are significant. Categorising relations into types: symmetric, asymmetric (both directions exist, with different effect sizes), and unidirectional (only one direction exists).
 - **RQ3.1:** Differences between symmetric and asymmetric models, with regards to intervention strategy (which intervention to do) and effectiveness (magnitude of change compared to the no-intervention case).
- **Individual heterogeneity in belief systems and intervention dynamics**
 - Second results section
 - **RQ2.2:** Fit models to conservative and liberal subsets of the data; compare and contrast. Compare with the model from the previous section, i.e., fit on the entire dataset.
 - **RQ3.2:**
 - Distribution of intervention effectiveness by individual.
 - How does intervention ranking vary across individuals?
 - Characterising how different initial states affect intervention success.
 - Looking at how other theory-driven features affect success, e.g.:
 - How receptive is the individual to the intervention?

The effective baseline ($h_i + \sum_j J_{ji}s_j$) determines the probability that $S_i^{t+1} = s$ for a state $s \in \pm 1$. Evaluating this after intervening provides a measurement for the success of the intervention on the intervention spin itself.

We could look at how this changes with different intervention strengths (it follows a logistic curve).

- How consistent is the individual’s belief state, as measured by the total system energy?
- How ‘entrenched’ is the target attitude?

Measure $h_k s_k + \sum_j J_{jk} s_j s_k$, where k is the target attitude.

- **General discussion**

- Draw out the key findings from the above two results sections
- Make clear the implications for our theory of belief system dynamics
- Sensitivity of parameter estimation to unmeasured factors or incorrect structural assumptions. i.e., what happens when we cannot include an influential belief, or when we falsely assume that a relation does or does not exist?
- Representational limitations of our model:
 - Pairwise relations limit what can be modelled. Demonstrate using vaccination example — we can’t capture relations between pairs of attitudes, whose effect sign or magnitude depends on a third belief/attitude.
 - Polar $(-1, +1)$ spin state assumption; some beliefs may be better treated as ‘on/off’, where the ‘off’ state has *no* effect on other spins, rather than a negative effect.
 - More generally, effect sizes may vary depending on specific spin states.

- **Dataset**

- Introduce the (insert name here) dataset, give context, survey details
- Validation, cleaning, transformations
- Question selection, indexes
- Binarisation

- **Literature review/related work**

- **Conclusions and future work**

Chapter 3

Non-equilibrium belief systems

3.1 Belief system dynamics

We will now formalise the dynamics of an individual belief system, under the following assumptions:

1. Beliefs and attitudes can be treated as discrete entities.
2. The state of a belief or attitude is Markovian with respect to the previous system configurations.
3. The probability distribution relating current and future belief states does not change over time.

The first assumption includes two key points. Firstly, that it makes sense to consider beliefs and attitudes as *entities*, or *objects*. This contrasts, for instance, alternative interpretations of attitudes as emissions from an underlying structure. (**TODO:** review CAN paper for further discussion on this matter). Secondly, that beliefs and attitudes can be considered *discrete* or *separable*. This is to say that there are no absolute physical constraints on the combinations of states which may be observed (although certain combinations may be highly unlikely). In particular, this captures the requirement that each belief or attitude can be represented using a single state variable.

Our second assumption concerns the treatment of beliefs and attitudes as Markovian.

- Harder to justify conceptually, since individuals have memory and can recall prior states beyond the previous second.

Consider a belief or attitude S_i with domain Ω_{S_i} . The state of S_i at time t depends on the previous states of other beliefs and attitudes, and we can therefore describe the trajectory of S_i as a Markov process:

$$\{s_i^t\}_{t=1}^{\infty} \quad \text{where} \quad s_i^t \in \Omega_{S_i} \quad (3.1)$$

with conditional state probability $P(S_i^{\tau+1} \mid \mathbf{S}^1, \dots, \mathbf{S}^{\tau}) = P(S_i^{\tau+1} \mid \mathbf{S}^{\tau})$ for $\tau \in \mathbb{N}$.

Since the state of S_i^{t+1} depends only on the previous belief system state, it follows that for any other belief or attitude S_j , the states S_i^{t+1} and S_j^{t+1} are conditionally independent given $\mathbf{S}^t$. We can therefore straightforwardly extend Equation 3.1 to describe the evolution of the entire belief system as a Markov process,

$$\{\mathbf{s}^t\}_{t=1}^{\infty} \quad \text{where} \quad \mathbf{s}^t \in \Omega_{S_1} \times \dots \times \Omega_{S_n} \quad (3.2)$$

with the probability of each belief system configuration given by the conditional distribution $P(\mathbf{S}^{t+1} \mid \mathbf{S}^t)$.

Under this conceptualisation, the task of modelling a belief system thus reduces to describing how the instantaneous configuration of belief states affects the probability distribution over possible future states.

3.2 Non-equilibrium belief system model

When studying the *symmetric* Ising model it is common to assume that the model is at equilibrium, such that $P(\mathbf{S}^t = \mathbf{s} \mid \mathbf{S}^{t-1}) = P(\mathbf{S} = \mathbf{s})$ for all configurations $\mathbf{s}$ and observation times t . The probability of observing $\mathbf{s}$ is then time-invariant and given by the Boltzmann probability parameterised by the Hamiltonian $H(\mathbf{s})$ (Cardy, 1996; Christensen & Moloney, 2005):

$$P(\mathbf{S} = \mathbf{s}) = \frac{e^{-\frac{1}{k_b T} \cdot H(\mathbf{s})}}{\sum_{\mathbf{s}'} e^{-\frac{1}{k_b T} \cdot H(\mathbf{s}')}} \quad (3.3)$$

where k_b is the Boltzmann constant, and the temperature parameter $T \in \mathbb{R}^+$ controls stochasticity. Equation 3.3 converges to a uniform distribution over all configurations when $T \rightarrow +\infty$, and a uniform distribution over the restricted set of configurations for which $H(\mathbf{s}')$ is minimised when $T \rightarrow 0^+$.

For the purposes of this study, we are interested in intervention dynamics, which are inherently non-equilibrium. To see why this is the case, notice that the purpose of intervention is to change the distribution of observed states. This is true both interventions intended to change the dominant state (e.g., to promote a sustainable alternative to a typical unsustainable behaviour) or reinforce it. Any belief system model intended for studying intervention dynamics therefore cannot assume equilibrium, and cannot define model dynamics using the Boltzmann distribution.

We instead define the conditional distribution $P(\mathbf{S}^t = \mathbf{s} \mid \mathbf{S}^{t-1})$ explicitly. Recall that the states of each pair of spins S_i, S_j at time t are conditionally independent random variables, given knowledge of the previous configuration $\mathbf{S}^{t-1}$ (Chapter 3.1). It therefore suffices for us to describe the distribution over spin states for an individual spin S_i at time t , as conditional on $\mathbf{S}^{t-1}$.

As in the standard Ising model formulation, we will assume that the model typically evolves in the direction of lower energy; however, rather than defining the energy at the level of the model using the Hamiltonian, we consider the energy experienced by S_i as a result of its baseline activation (also known as a *local field effect*), and interactions with other spins.

The baseline activation $h_i \in \mathbb{R}$ describes the tendency for S_i to take on the value $+1$ in the absence of interactions with other spins, or, more generally, when pairwise spin interactions have a net-effect of zero. S_i tends to adopt the value $+1$ under these circumstances, iff, $h_i > 0$, and -1 iff $h_i < 0$. This tendency increases with the magnitude of h_i .

Other spins influence S_i 's state through alignment or opposition relations. For a spin S_j , we define an interaction effect J_{ji} (read '*influence of j on i* '). When J_{ij} is positive or negative, S_i is influenced to adopt the state S_j^{t-1} or $-S_j^{t-1}$ respectively. In the special case when $j = i$, we refer to J_{ii} as a *self-influence* effect. Positive self-influence effects reflect the inertia of S_i , i.e., the tendency to sustain a particular belief or attitude irrespective of other beliefs and attitudes. Negative self-influence effects are not clearly interpretable. The inclusion of self-influence effects allows us to capture the timescales of different beliefs or attitudes independently.

The energy experienced by S_i for a particular spin state $s \in \{-1, +1\}$ is then the result of s in combination with the baseline activation and influence effects from all spins with edges to S_i :

$$H_i(s \mid \mathbf{s}^{t-1}) = h_i \cdot s + \sum_j A_{ji} J_{ji} s_j^{t-1} \cdot s \quad (3.4)$$

where $\mathbf{A} \in \{0, 1\}^{n \times n}$ is a pairwise adjacency matrix, with $A_{ji} = 1$, iff, S_j influences S_i . Dividing the common factor of s , we can equivalently write Equation 3.4 as

$$H_i(s \mid \mathbf{s}^{t-1}) = s \cdot h_i^{\text{eff}}(\mathbf{s}^{t-1}) \quad (3.5)$$

where $h_i^{\text{eff}}(\mathbf{s}^{t-1}) = h_i + \sum_j A_{ji} J_{ji} s_j^{t-1}$ is the effective baseline activation experienced by S_i at time t . We then define the probability that S_i adopts the state s at time t as:

$$\begin{aligned} P(S_i^t = s \mid \mathbf{S}^{t-1} = \mathbf{s}) &= \frac{e^{\frac{1}{T} H_i(s \mid \mathbf{s})}}{\sum_{s' \in \{0, 1\}} e^{\frac{1}{T} H_i(s' \mid \mathbf{s})}} \\ &= \frac{e^{\frac{1}{T} H_i(s \mid \mathbf{s})}}{e^{\frac{1}{T} H_i(s \mid \mathbf{s})} + e^{-\frac{1}{T} H_i(-s \mid \mathbf{s})}} \\ &= \frac{e^{\frac{1}{T} s \cdot h_i^{\text{eff}}(\mathbf{s})}}{e^{\frac{1}{T} s \cdot h_i^{\text{eff}}(\mathbf{s})} + e^{-\frac{1}{T} s \cdot h_i^{\text{eff}}(\mathbf{s})}} \end{aligned} \quad (3.6)$$

For $s = +1$, this reduces to the logistic function:

$$\begin{aligned} p_i^s &:= P(S_i^t = +1 \mid \mathbf{S}^{t-1} = \mathbf{s}) = \frac{e^{\frac{1}{T} \cdot h_i^{\text{eff}}(\mathbf{s})}}{e^{\frac{1}{T} \cdot h_i^{\text{eff}}(\mathbf{s})} + e^{-\frac{1}{T} \cdot h_i^{\text{eff}}(\mathbf{s})}} \\ &= \frac{1}{1 + e^{-2 \cdot \frac{1}{T} \cdot h_i^{\text{eff}}(\mathbf{s})}} \\ &= \text{logistic}(2h_i^{\text{eff}}(\mathbf{s})/T) \end{aligned} \quad (3.7)$$

Therefore the complete conditional distribution is given by:

$$\begin{aligned} P(\mathbf{S}^t = \mathbf{s}^t \mid \mathbf{S}^{t-1} = \mathbf{s}^{t-1}) &= \prod_{i=1}^n P(S_i^t = s_i \mid \mathbf{S}^{t-1} = \mathbf{s}^{t-1}) \\ &= \prod_{i=1}^n \left[\left(\frac{1 + s_i}{2} \right) p_i^{s_i^{t-1}} + \left(\frac{1 - s_i}{2} \right) (1 - p_i^{s_i^{t-1}}) \right] \end{aligned} \quad (3.8)$$

NOTE: The initial distribution must be specified, since the model only captures transition probabilities.

3.3 Symmetric and asymmetric belief systems

Let $\mathcal{M}$ be a belief system model defined as in the preceding section, comprising $N \in \mathbb{N}$ beliefs or attitudes, with adjacency matrix $\mathbf{A} \in \{0, 1\}^{N \times N}$, interaction effect matrix $\mathbf{J} \in \mathbb{R}^{N \times N}$, and baseline activations $\mathbf{h} \in \mathbb{R}^N$.

When each entry in $\mathbf{J}$ is independent of all others, we say that $\mathcal{M}$ is an **asymmetric belief system**. The term *asymmetric* here refers to the directed relations between a pair of nodes. In an asymmetric belief system, for any pair of distinct nodes $S_i \neq S_j$, it may be the case that an influence relation exists only in one direction, or that the directed relations differ in magnitude.

In the special case where we constrain $A_{ij} = A_{ji}$ and $J_{ij} = J_{ji}$ for all $i, j \in [1, N]$, we instead say that $\mathcal{M}$ is a **symmetric belief system**. In a symmetric belief system all interactions are directionally-equivalent. This can be considered a simple extension of the symmetric Ising model with (i) self-interaction terms and (ii) temporal dynamics, thus the symmetric belief system model tends toward an equilibrium steady state.

3.4 Simulation via Glauber dynamics

It is straightforward to draw samples from a belief system model using Glauber dynamics, given the conditional probability distribution defined in Equation 3.8.

Given an initial state $\mathbf{s}^0 \in \{-1, +1\}^N$, we sample a sequence of $T \in \mathbb{N}$ subsequent configurations:

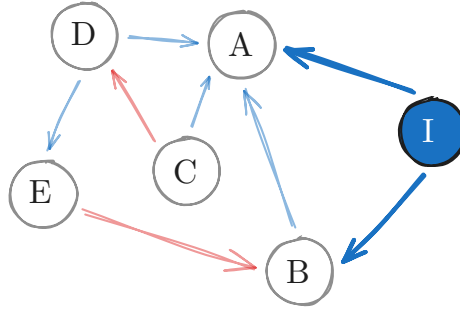
$$\{\mathbf{s}^t\}_{t=1}^T, \quad \text{where each } \mathbf{s}^t \sim P(\mathbf{S}^t \mid \mathbf{S}^{t-1}) \quad (3.9)$$

Each belief or attitude has the opportunity to update during every time interval. This update routine is referred to as *synchronous* Glauber dynamics, contrasting *asynchronous* Glauber dynamics, in which only one spin can update during a given interval.

3.5 Modelling interventions

We now outline our approach to modelling interventions in the asymmetric belief system model. In this study we consider interventions which affect the *state* of a belief system. Notably, this excludes interventions intended to influence the structure of a belief system, for instance, by changing the existence or effect size of influence relations between beliefs and attitudes.

Let $\mathcal{M}$ be a belief system model with parameters $\langle \mathbf{A}, \mathbf{J}, \mathbf{h} \rangle$. We can consider an intervention as an auxiliary node, I , in the belief system network, with state fixed at a particular value and outgoing edges toward a subset of beliefs and attitudes:



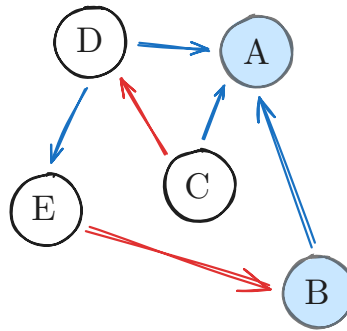
The intervention node I exerts influence on the belief system nodes A and B . Let us consider the effective baseline activation at node A and time $t + 1$, as defined in Equation 3.4, given a previous configuration $\mathbf{s}^t$:

$$h_A^{\text{eff}}(\mathbf{s}^t) = h_A + \sum_{\text{node } j} A_{jA} J_{jA} s_j^t \quad (3.10)$$

Since the state of I is fixed (in this case, at +1), the contribution of the intervention node to A 's effective activation baseline is constant. We can then re-write Equation 3.10, interpreting the intervention effect as an adjustment to the baseline activation h_A :

$$h_A^{\text{eff}}(\mathbf{s}^t) = (h_A + J_{IA}) + \sum_{\text{node } j \neq I} A_{jA} J_{jA} s_j^t \quad (3.11)$$

It follows then that we can model interventions more simply as an adjustment to the baseline activation of certain beliefs and attitudes:



Let us define this more formally. For a belief system model $\mathcal{M}$ with $N \in \mathbb{N}$ nodes, and parameters $\langle \mathbf{A}, \mathbf{J}, \mathbf{h} \rangle$, an *intervention*, is the function $\varphi_{\mathcal{M}} : \delta_h \mapsto \mathcal{M}'$, where $\delta_h \in \mathbb{R}^N$ is a vector of offsets to the baseline activations, and the model $\mathcal{M}'$ has parameters $\langle \mathbf{A}, \mathbf{J}, \mathbf{h}' \rangle$, where

$$\mathbf{h}' = \mathbf{h} + \delta_h \quad (3.12)$$

TODO: Include discussion on the choice of δ_h , with reference to a figure showing how the activation probability depends on h_i^{eff} .

Chapter 4

Methods

4.1 To-do

- Parameter estimation: Discussion on why we can't estimate from cross-sectional data, even when excluding self-loops.
 - If we could, this would simplify dataset. Wouldn't need to rely on repeating participants.

4.2 Counterfactual interventions

Individuals have different belief systems. Influence relations between beliefs and attitudes reflect heterogeneity in individual experiences and perception. Often these relations can themselves be considered beliefs — whether or not support for climate change mitigation leads to support for any specific policy depends in-part on an individual's belief regarding the policy's relevance. An individual's baseline activations, in turn, reflect a culmination of factors which determine their tendency toward certain beliefs or attitudes.

Our approach to parameter estimation, however, fundamentally assumes that the inferred belief system is shared by all individuals in the dataset from which the model is estimated. We note that this is not strictly a limitation of the parameter estimation method, nor of the model itself. Rather this assumption is imposed by data limitations. Fitting individual models requires substantial data from each individual, which is not available in the climate attitudes dataset. Moreover, even with substantial observational data, we are unlikely to observe all state transitions necessary to understand the

potential dynamics of an individual system, owing to the relatively slow-moving nature of beliefs and attitudes.

For any given individual, the inferred models may thus not accurately reflect the stability of belief system states. A configuration which is stable in a given individual's own belief system may be considered unstable in the shared model. When simulating interventions, we take the most recent measurement for each individual as their initial state. However, due to this perceived instability, we expect to observe natural dynamics away from this initial state, even in null-intervention scenarios.

We thus consider intervention effects as relative to the null-intervention counterfactual baseline rather than relative to the initial state. For each intervention we run both intervention and null ($\delta_h = 0$) scenarios with identical random number generation settings, and define the *intervention effect* as the difference between the observed outcomes.

4.3 Parameter estimation

For the purposes of the experiments in the following sections, we perform parameter estimation to calibrate the belief system model described in Chapter 3.2 to the climate attitudes dataset (Chapter 8). We first outline the context of the parameter estimation problem, and then present our approach. We omit derivation details in this section; however, the curious reader can find these in Appendix A (**TODO**).

Consider a population of $M \in \mathbb{N}$ individuals with a shared belief system $\mathcal{M}$ comprising $N \in \mathbb{N}$ beliefs or attitudes with pre-defined adjacency matrix $\mathbf{A} \in \{0, 1\}^{N \times N}$, for which the parameters (i.e., the baseline activations and interaction effects) are unknown. Suppose that for each individual $m \in [1, M]$ we have observed a series of measurements, reflecting that individual's belief system state at each of $t \in [1, T]$ uniformly spaced timesteps:

$$\{\mathbf{x}_{(m)}^t\}_{t=1}^T, \quad \text{where each } \mathbf{x}_{(m)}^t \in \mathbb{R}^N \quad (4.1)$$

The measurements need not be binary, but are assumed to be real. The collection of measurements across all individuals forms a dataset D , which is the particular observed value of the random variable $\mathcal{D}$ representing possible datasets.

While maximum likelihood estimation (MLE) is often used to infer parameters for similar models (Lee et al., 2015; Nguyen et al., 2017), we should be cautious, for two reasons, of applying this method naïvely in the present study.

Firstly, the model defined in Equation 3.8 assumes binary spin states $s \in \{-1, +1\}$. The climate attitudes dataset does not satisfy this assumption and must be binarised for MLE to be applicable. However, we have no guarantee that the parameterisation inferred for any specific binarisation is a reasonable explanation for the non-binarised dataset, nor for any other possible binarisation in the case where a probabilistic binarisation scheme is used (such as the one described in Chapter 8.4).

- Formally, can say that any specific binarisation $D_B \sim \text{Bin}(D)$ and model $\mathcal{M}$ estimated from D_B :

$$I(D; \mathcal{M}) \leq I(D_B; \mathcal{M}) \quad (4.2)$$

- And this inequality is strict when $\text{Bin}(D)$ is not constant.

Secondly, maximum likelihood estimation is prone to overfitting when the number of model parameters is similar to the number of observations (Epskamp, Borsboom, et al., 2018). While we consider only a small number of beliefs and attitudes in this study ($N \in \{7, 8\}$), the number of parameters p grows rapidly, with $p \in O(N^2)$ for both the symmetric and asymmetric model variants. In combination with sampling error, we expect the number of parameters to negatively impact the robustness of the inferred parameters — given an alternative dataset of the same size, we would expect significant variation in parameters.

We mitigate these respective issues in our parameter estimation scheme by (i) marginalising over the set of possible binarisations of D , and (ii) applying regularisation based on parameter magnitudes. We choose the parameterisation $\theta^* \in \mathbb{R}^p$ which satisfies the optimisation problem:

$$\theta^* = \underset{\theta \in \mathbb{R}^p}{\text{argmax}} f(D; \hat{\theta}), \quad \text{where} \quad f(D; \hat{\theta}) := \mathcal{L}_D(\hat{\theta}) - \lambda \sum_{\theta \in \hat{\theta}} \sqrt{\theta^2 + \varepsilon} \quad (4.3)$$

The first term in the objective function f , $\mathcal{L}_D(\hat{\theta})$, denotes the *expected* log-likelihood given a parameterisation $\hat{\theta}$, over possible binarisations of D :

$$\mathcal{L}_D(\hat{\theta}) = \mathbb{E}[L_{D_B}(\hat{\theta}) \mid \mathcal{D} = D] = \sum_{D_B} P(\text{Bin}(\mathcal{D}) = D_B \mid \mathcal{D} = D) \cdot L_{D_B}(\hat{\theta}) \quad (4.4)$$

where $L_{D_B}(\hat{\theta})$ is the log-likelihood of a specific binarisation D_B . We will defer explicitly defining $L_{D_B}(\hat{\theta})$ for now, but will return to this point shortly in Equation 4.5.

TODO: Intuition for the effect of using the expectation.

- Inferred model reflects ambiguity in states close to zero.
- The likelihood contribution of each data value is the expectation over possible spin states.
- Values close to zero contribute a comparable weighting from each binary state.
- When the likelihood given one binary state would be significantly higher than for the other, this is downweighted to account for the probability that the value does not obtain this state.
- Therefore the expectation prevents overconfident optimisation toward any specific binarisation — the inferred model is required to explain the binarised dataset *in expectation*.
- In other words, if a data value may be binarised to -1 or $+1$ with equal probability, then both cases should be reasonably explained by the parameterisation.

The second term is a smooth variant of L1 regularisation (Tibshirani, 1996). This has the effect of penalising nonzero parameters which do not meaningfully contribute to the expected likelihood. Parameters instead must reflect relationships which are sufficiently prevalent in the data, reducing the robustness issues resulting from sampling error and a large number of parameters. The hyperparameter $\lambda \in \mathbb{R}^+$ controls regularisation strength, with higher values resulting in sparser models. As $\varepsilon \rightarrow 0^+$ the second term converges to the standard formulation of L1 regularisation. Unlike the standard formulation, when $\varepsilon > 0$ the first-derivative of this term exists, so is amenable to gradient-based optimisation. We discuss the choice of values for λ and ε in Chapter 4.3.1.

Let $D_B \sim \text{Bin}(D)$ be a possible binarisation of D , and θ a parameterisation for the (asymmetric or symmetric) belief system model, which is decomposable into the model parameters $\theta = \langle \mathbf{J}, \mathbf{h} \rangle$. The log-likelihood of D_B given θ is:

$$\begin{aligned} L_{D_B}(\theta) &= \log P(D_B \mid \theta) \\ &= \sum_{m=1}^M \sum_{t=1}^{T-1} \log s_{i,(m)}^{t+1} \cdot h_i^{\text{eff}}(\mathbf{s}_{(m)}^t) - \log(2 \cosh h_i^{\text{eff}}(\mathbf{s}_{(m)}^t)) \end{aligned} \quad (4.5)$$

The derivation of Equation 4.5 is analogous to that of the non-equilibrium Ising model log-likelihood (Nguyen et al., 2017). Combining Equation 4.5 and Equation 4.4, we obtain an explicit expression for the expected likelihood:

$$\mathcal{L}_D(\boldsymbol{\theta}) = \sum_{\mathbf{s}} \left(\sum_{m=1}^M \sum_{t=1}^{T-1} P(\mathbf{S}_{(m)}^t = \mathbf{s} \mid D) \cdot \mathbb{E}[\mathbf{S}_{(m)}^{t+1} \mid D]^T h^{\text{eff}}(\mathbf{s}) \right) - Z(D; \boldsymbol{\theta}) \quad (4.6)$$

where $\mathbb{E}[\mathbf{S}_{(m)}^{t+1} \mid D]$ is the expected binary configuration for the observation $\mathbf{x}_{(m)}^{t+1}$, the vector $h^{\text{eff}}(\mathbf{s})$ contains the effective baseline activation for each spin given the previous state $\mathbf{s}$, and Z is the expected log partition function, summed across observations:

$$Z(D; \boldsymbol{\theta}) = \sum_{\mathbf{s}} \sum_{\substack{m \leq M \\ t < T}} P(\mathbf{S}_{(m)}^t = \mathbf{s} \mid D) \sum_{i=1}^N \log(2 \cosh h_i^{\text{eff}}(\mathbf{s})) \quad (4.7)$$

Per the optimisation problem defined in Equation 4.3, we choose the parameterisation $\boldsymbol{\theta}^*$ which maximises the regularised expected log-likelihood, which occurs when $\frac{\partial f}{\partial \boldsymbol{\theta}} = 0$ for every parameter $\boldsymbol{\theta} \in \boldsymbol{\theta}^*$. The partial derivative with respect to a given baseline activation parameter h_i , for $i \in [1, N]$, is given by:

$$\begin{aligned} \frac{\partial}{\partial h_i} f(D; \boldsymbol{\theta}) &= \sum_{\mathbf{s}} \sum_{\substack{m \leq M \\ t < T}} P(\mathbf{S}_{(m)}^t = \mathbf{s} \mid D) [\mathbb{E}[S_{i,(m)}^{t+1} \mid D] - \tanh(h_i^{\text{eff}}(\mathbf{s}))] \\ &\quad - \frac{\lambda h_i}{\sqrt{h_i^2 + \varepsilon}} \end{aligned} \quad (4.8)$$

Note that the expressions for $\mathcal{L}_D(\boldsymbol{\theta})$ and $\frac{\partial}{\partial h_i} f(D; \boldsymbol{\theta})$ are applicable to both the asymmetric and symmetric belief system models. This property does not hold for the partial derivatives with respect to interaction effects. In the asymmetric model, for $i, j \in [1, N]$, this is derived analogously to Equation 4.8 as:

$$\begin{aligned} \frac{\partial}{\partial J_{ji}} f(D; \boldsymbol{\theta}) &= \sum_{\mathbf{s}} \sum_{\substack{m \leq M \\ t < T}} P(\mathbf{S}_{(m)}^t = \mathbf{s} \mid D) A_{ji} s_j [\mathbb{E}[S_{i,(m)}^{t+1} \mid D] - \tanh(h_i^{\text{eff}}(\mathbf{s}))] \\ &\quad - \frac{\lambda J_{ji}}{\sqrt{J_{ji}^2 + \varepsilon}} \end{aligned} \quad (4.9)$$

In the symmetric belief system model, since we require that $\mathbf{J} = \mathbf{J}^T$, we estimate a single interaction parameter $J_{ji} = J_{ij} = \beta_{\{i,j\}}$ for each pair of spins S_i, S_j . Each pair thus contributes twice to the partial derivative, once for each direction:

Parameter	Model type	Dataset	Value
λ	Symmetric	Full	1.4×10^{-2}
	Asymmetric	Full	1.2×10^{-2}
		Conservative	3.4×10^{-2}
		Liberal	4.3×10^{-2}
ε	All	All	10^{-8}

Table 4.1: Values for regularisation strength (λ) and smoothing (ε) hyperparameters. Regularisation strength model- and dataset-specific; *Conservative* and *Liberal* refer to subsets of the climate attitudes dataset (**TODO**: reference) comprising individuals with the specified ideology.

$$\begin{aligned} \frac{\partial}{\partial \beta_{\{i,j\}}} f(D; \theta) = & \sum_{\mathbf{s}} \sum_{\substack{m \leq M \\ t < T}} P(\mathbf{S}_{(m)}^t = \mathbf{s} \mid D) \sum_{(i', j') \in \{i,j\}} A_{j' i'} s_{j'} [\mathbb{E}[S_{i', (m)}^{t+1} \mid D] - \tanh(h_{i'}^{\text{eff}}(\mathbf{s}))] \\ & - \frac{\lambda \beta_{\{i,j\}}}{\sqrt{\beta_{\{i,j\}}^2 + \varepsilon}} \end{aligned} \quad (4.10)$$

4.3.1 Hyperparameters

The optimisation problem in Equation 4.3 has two hyperparameters which must be specified prior to parameter estimation: the regularisation strength $\lambda \in \mathbb{R}^+$, and the smoothing parameter $\varepsilon \in \mathbb{R}^+$. The values for both are summarised in Table 4.1.

We use the Extended Bayesian Information Criterion (EBIC) (Chen & Chen, 2008) to select λ , as recommended by Epskamp, Borsboom, et al. (2018):

$$\text{EBIC}(\theta^*) = k \cdot [\ln(M(T-1)) + 2\gamma \ln(p)] - 2\mathcal{L}_D(\theta^*) \quad (4.11)$$

The variable k denotes the number of nonzero parameters in the optimised model, where a parameter θ is considered nonzero if $|\theta| > 10^{-2}$. We take $\gamma = 0.25$, which has been shown to work well in general for network inference in the inverse Ising problem (Barber & Drton, 2015).

The EBIC is an evaluative criterion for model selection, which balances maximisation of the log-likelihood with model complexity, as measured by the number of parameters. Compared to the original Bayesian Information Criterion, EBIC tends to be more conservative when the number of parameters and observations are comparable (Foygel & Drton, 2010). We select λ which minimises the EBIC for the symmetric and asymmetric models (independently), the results of which are displayed in Figure 4.1.

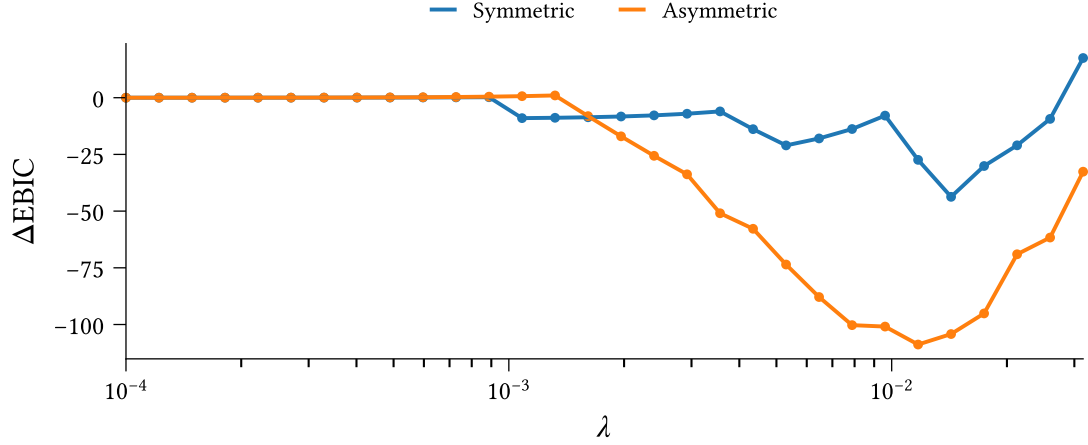


Figure 4.1: Effect of regularisation strength λ on Extended Bayesian Information Criterion for symmetric and asymmetric belief system models optimised to the climate attitudes dataset using Equation 4.3. The vertical axis measures the difference in EBIC compared to the no-regularisation case ($\lambda = 0$).

We consider parameters with magnitude less than 10^{-2} as ‘effectively zero’, and take $\varepsilon = 10^{-8}$ such that $\sqrt{\varepsilon}$ is significantly smaller than this threshold. This choice of ε ensures that the smooth regularisation remains a good approximation to L1 regularisation, i.e.,

$$\sqrt{\theta^2 + \varepsilon} \approx |\theta| \quad (4.12)$$

4.3.2 Implementation details

We solve the parameter estimation problem using the Scipy 1.17.1 implementation of the quasi-Newton BFGS optimisation algorithm (Nocedal & Wright, 2006, p. 136; Virtanen et al., 2020). We use the analytic jacobian comprising the partial derivatives stated above. We take $\boldsymbol{\theta} = \mathbf{0}$ as the initial guess.

To ensure numerical stability irrespective of dataset size, we rescale the expected log-likelihood contribution to the objective function and partial derivatives, dividing by the number of observed observations (time intervals) in the dataset. For $M \in \mathbb{N}$ individuals and $T \in \mathbb{N}$ timesteps this amounts to a scale factor of $\frac{1}{M(T-1)}$.

Chapter 5

Existence and impact of asymmetry in belief systems

5.1 Plan

First results section, for experiments on the effect of asymmetry

- **RF1:** Show asymmetric model fit
- **RQ2.1:** Existence of asymmetric relations. Some, but not all, are significant. Categorising relations into types: symmetric, asymmetric (both directions exist, with different effect sizes), and unidirectional (only one direction exists).
- **RQ3.1:** Differences between symmetric and asymmetric models, with regards to intervention strategy (which intervention to do) and effectiveness (magnitude of change compared to the no-intervention case).

5.2 Fitting model

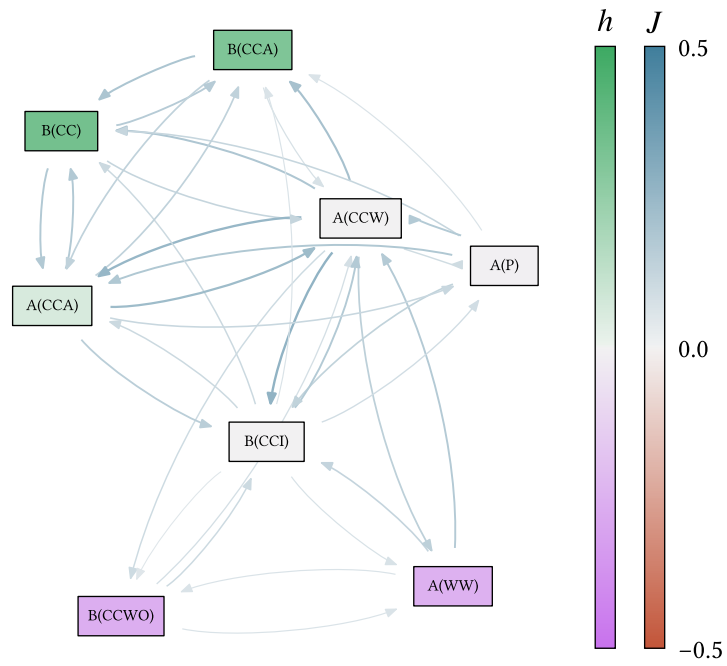


Figure 5.1: TODO

5.3 RQ2.1: Asymmetry in estimated models

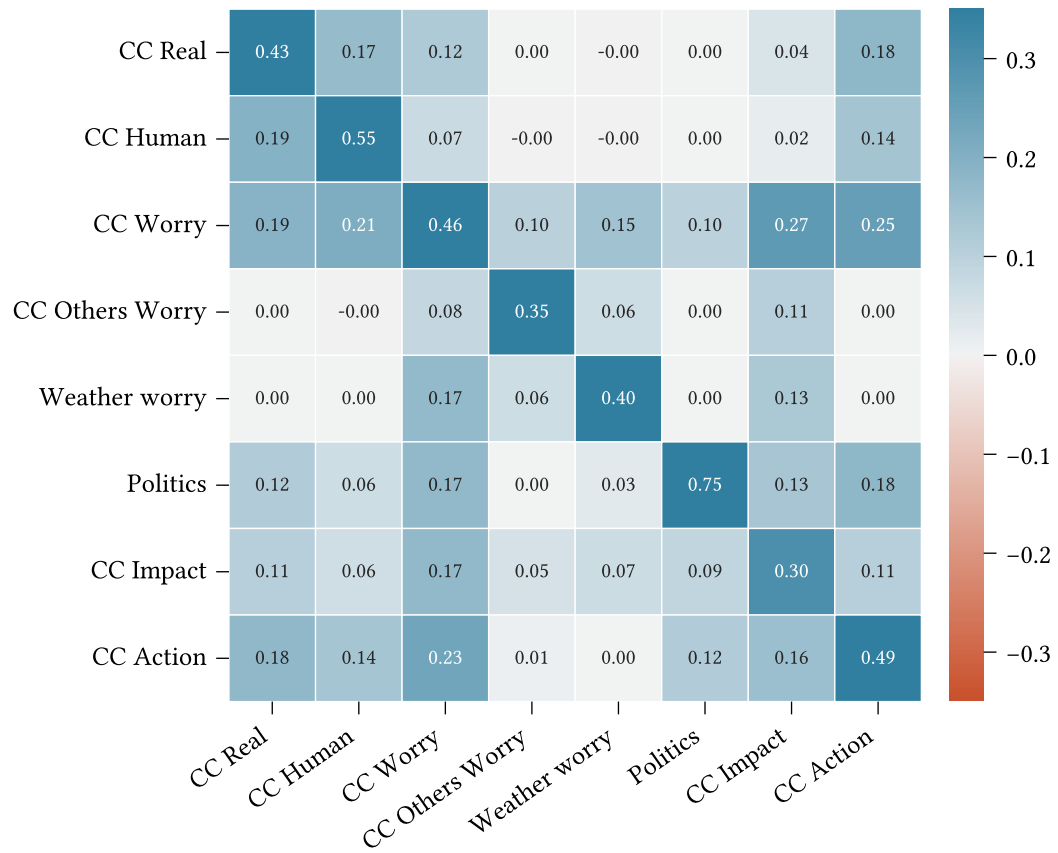


Figure 5.2: Climate belief system interaction effect matrix.

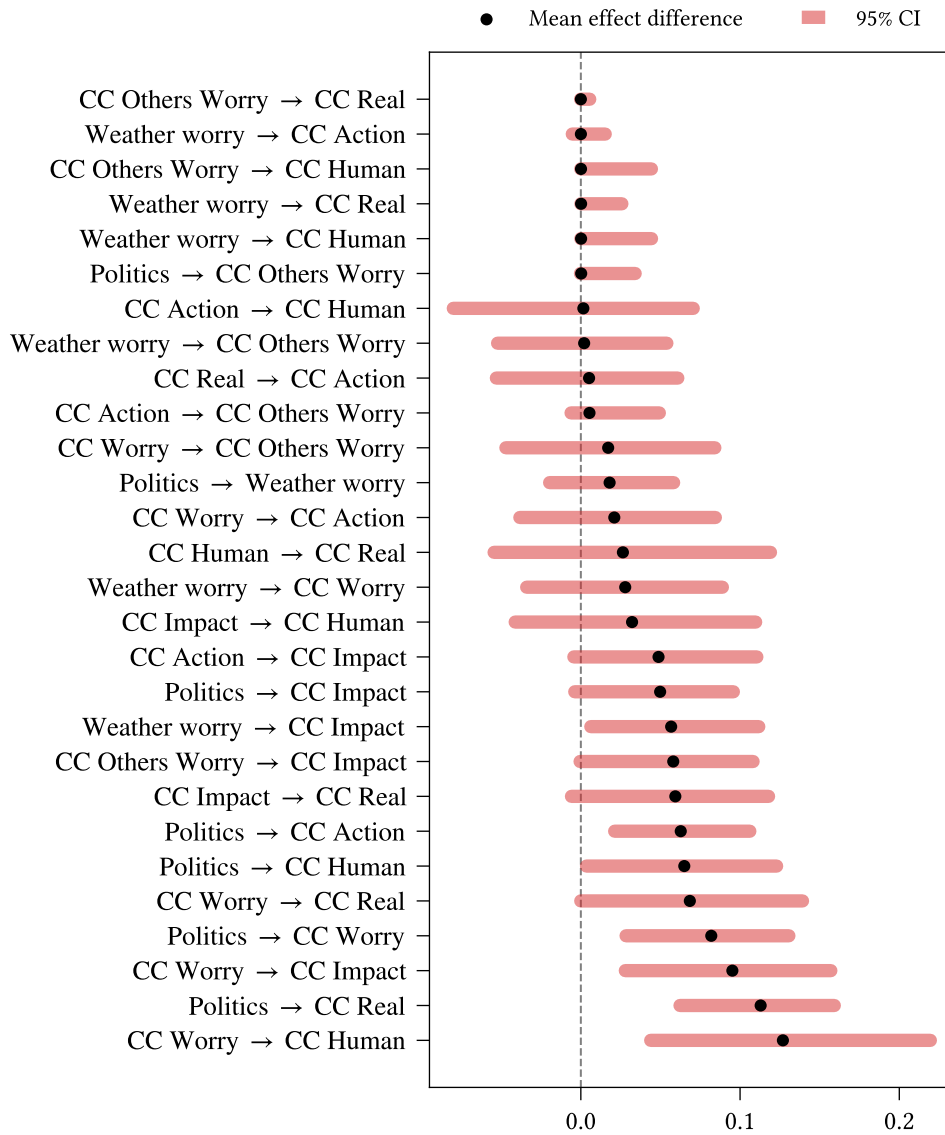


Figure 5.3: TODO

5.4 RQ3.1: Intervention strategy and effectiveness under sym/asym assumptions

Fit models using bootstrapping. Consider both effects of intervention (how does intervening at S affect other spins) and strategies (how does intervening at different spins affect this *one*). Repeat for varying levels/strengths of intervention.

For strategy, estimate expected ranking of interventions at both collective and individual levels. For the former, measure collective effect, then rank interventions, then take mean

across repeats. For the latter, measure expected individual effect, rank interventions, then take mean across individuals.

5.4.1 Ranking interventions

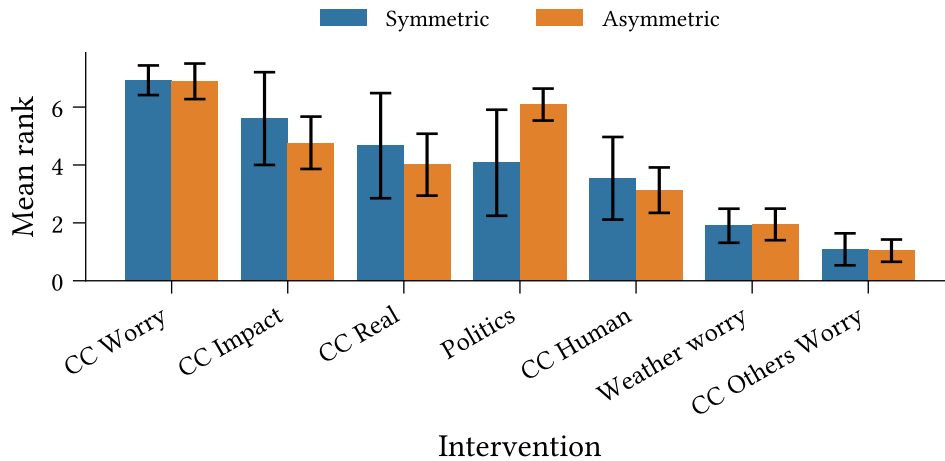


Figure 5.4: Expected ranking for weak interventions ($\delta h_i = 0.5$) targeting individuals' support for climate-related policies. A higher rank indicates that an intervention is more effective, as measured by the proportion of the population who support climate policy after 30 months in simulation time.

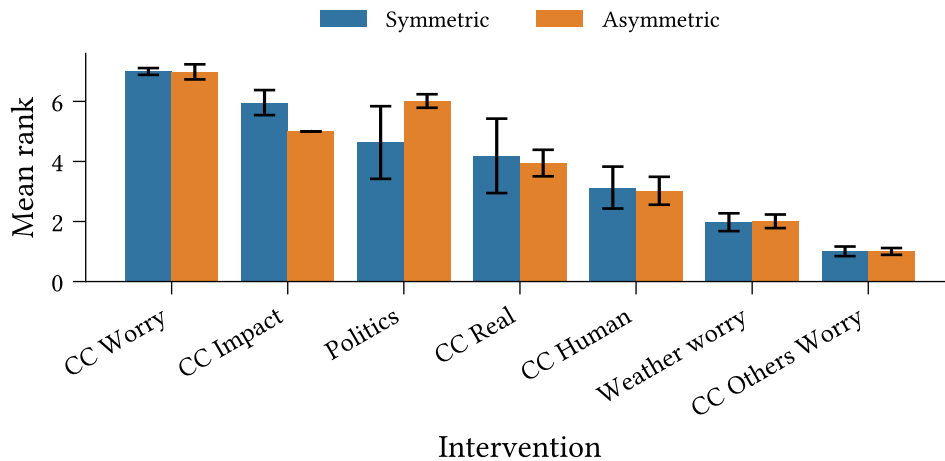


Figure 5.5: Expected ranking for medium interventions ($\delta h_i = 1.0$) targeting individuals' support for climate-related policies. A higher rank indicates that an intervention is more effective, as measured by the proportion of the population who support climate policy after 30 months in simulation time.

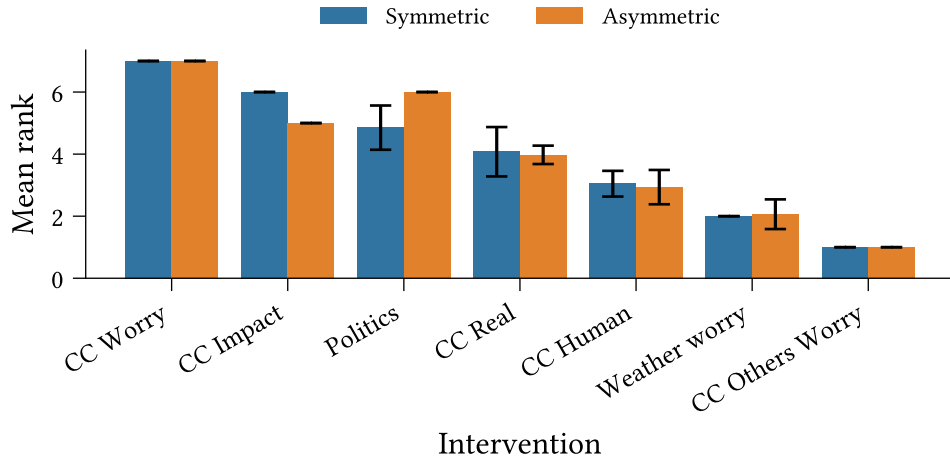


Figure 5.6: Expected ranking for perfect interventions ($\text{do}(S_i = +1)$) targeting individuals' support for climate-related policies. A higher rank indicates that an intervention is more effective, as measured by the proportion of the population who support climate policy after 30 months in simulation time.

5.4.2 Intervention effect

Measures the average difference in state between intervention and no-intervention scenarios

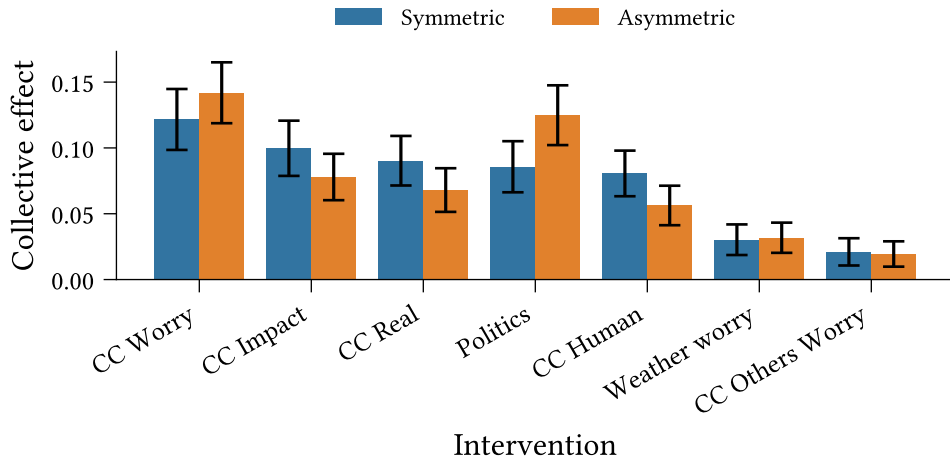


Figure 5.7: Expected effect of weak interventions ($\delta h_i = 0.5$) targeting individuals' support for climate-related policies. Effect is calculated as the difference in state after 30 months, as observed in the intervention scenario, compared to the no-intervention scenario.

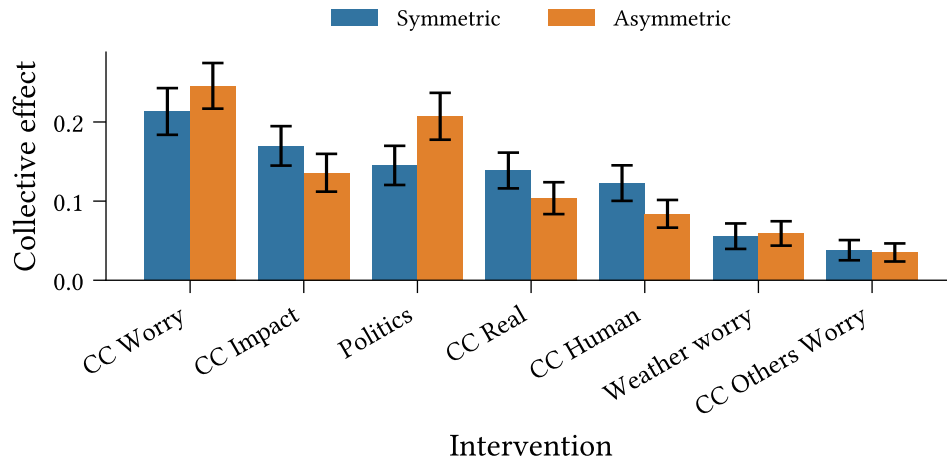


Figure 5.8: Expected effect of medium interventions ($\delta h_i = 1.0$) targeting individuals' support for climate-related policies. Effect is calculated as the difference in state after 30 months, as observed in the intervention scenario, compared to the no-intervention scenario.

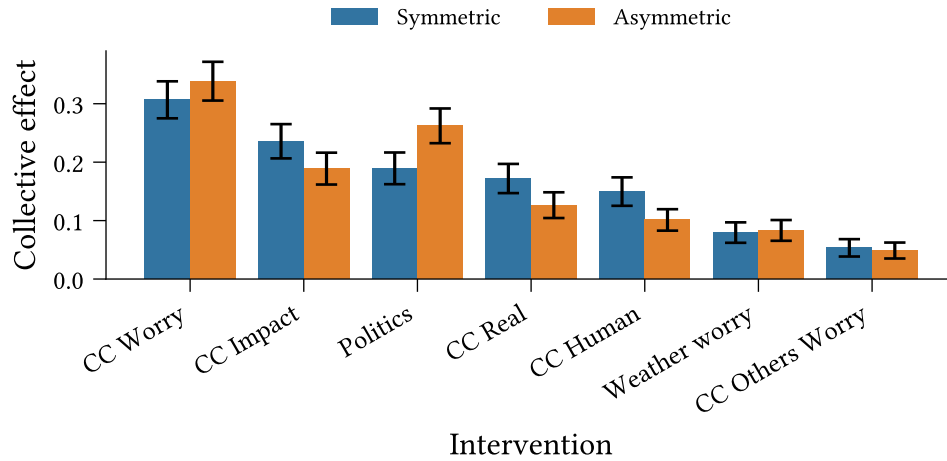


Figure 5.9: Expected effect of perfect interventions ($\text{do}(S_i = +1)$) targeting individuals' support for climate-related policies. Effect is calculated as the difference in state after 30 months, as observed in the intervention scenario, compared to the no-intervention scenario.

Chapter 6

Heterogeneous belief systems and intervention effects

6.1 Plan

Second results section, on individual heterogeneity in belief system structure, and intervention dynamics.

- **RQ2.2:** Fit models to conservative and liberal subsets of the data; compare and contrast. Compare with the model from the previous section, i.e., fit on the entire dataset.
- **RQ3.2:**
 - Distribution of intervention effectiveness by individual.
 - How does intervention ranking vary across individuals?
 - Characterising how different initial states affect intervention success.
 - Looking at how other theory-driven features affect success, e.g.:
 - How receptive is the individual to the intervention?

The effective baseline ($h_i + \sum_j J_{ji}s_j$) determines the probability that $S_i^{t+1} = s$ for a state $s \in \pm 1$. Evaluating this after intervening provides a measurement for the success of the intervention on the intervention spin itself.

We could look at how this changes with different intervention strengths (it follows a logistic curve).

- How consistent is the individual's belief state, as measured by the total system energy?

- How ‘entrenched’ is the target attitude?

Measure $h_k s_k + \sum_j J_{jk} s_j s_k$, where k is the target attitude.

6.2 RQ2.2: differences between belief systems for different (types of) individuals

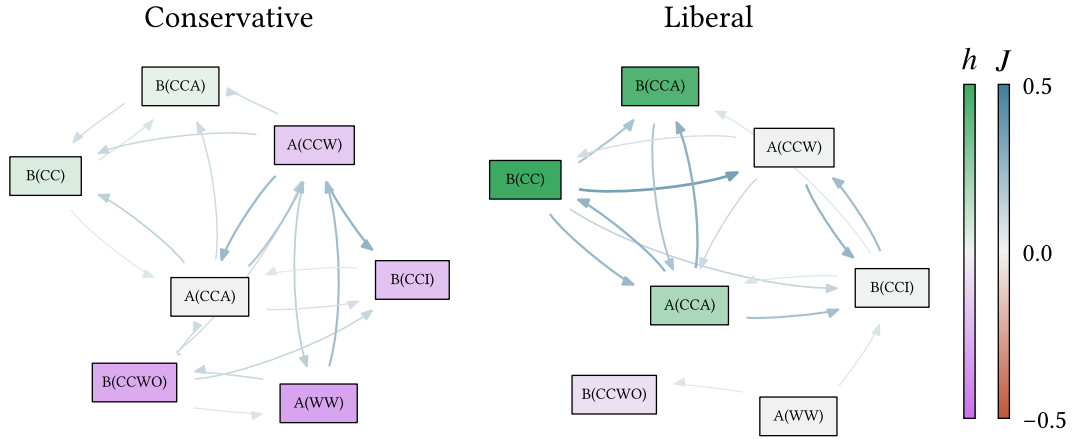


Figure 6.1: Prefixes: A (attitude), B (belief); node labels: CC (climate change), CCA (climate change anthropogenic), CCW (climate change worry), CCWO (climate change worry others), CCI (climate change impacts), CCP (climate change policies), WW (weather worry).

6.3 RQ3.2: Intervention strategy and effectiveness across individuals

Examine distributions of expected outcome effects (per individual) for different interventions. Characterise the individuals found at different parts of the distribution (personas, average magnetisation).

6.3.1 Intervention ranking

For each individual, compute ranking over the expected effects of each possible intervention.

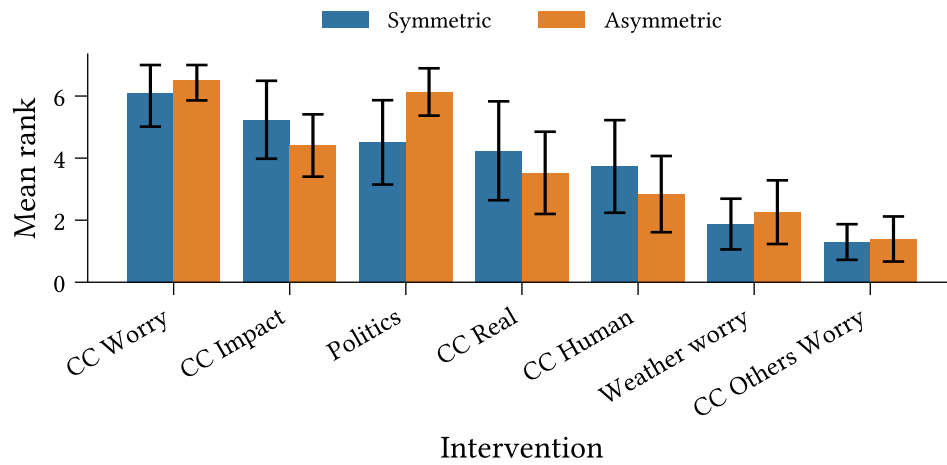


Figure 6.2: Ranked intervention effects per-individual, weak, targeting climate policy

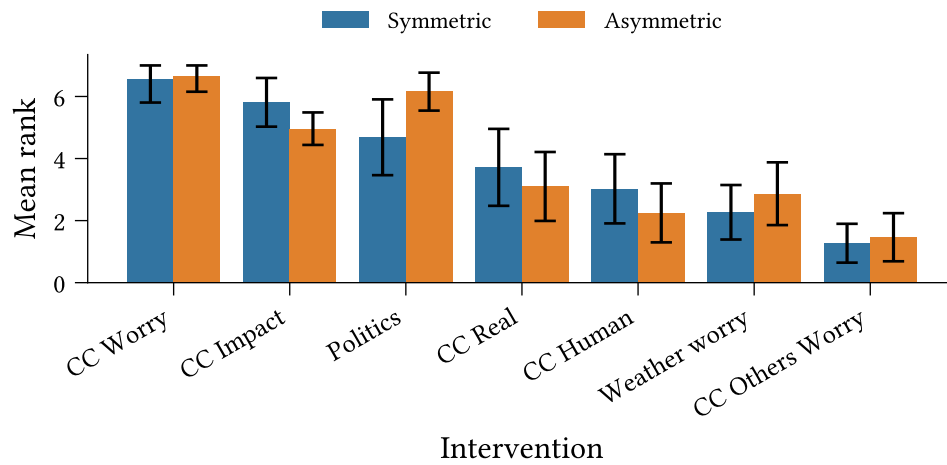


Figure 6.3: Ranked intervention effects per-individual, perfect, targeting climate policy

6.3.2 Who is intervention effective for?

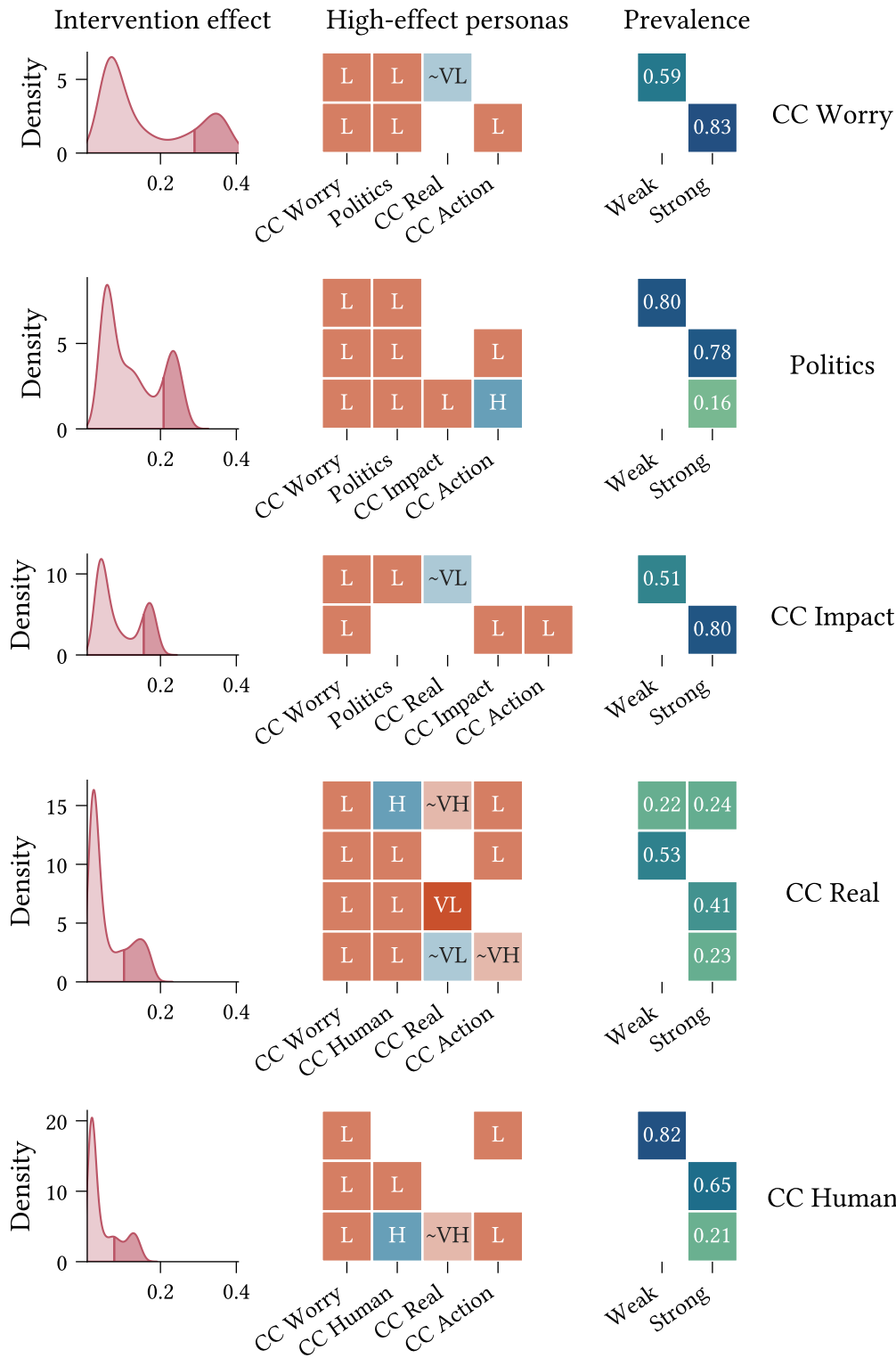


Figure 6.4: **TODO**

Chapter 7

Discussion

7.1 Plan

- Draw out the key findings from the above two results sections
- Make clear the implications for our theory of belief system dynamics
- Sensitivity of parameter estimation to unmeasured factors or incorrect structural assumptions. i.e., what happens when we cannot include an influential belief, or when we falsely assume that a relation does or does not exist?
- Representational limitations of our model:
 - Pairwise relations limit what can be modelled. Demonstrate using vaccination example — we can't capture relations between pairs of attitudes, whose effect sign or magnitude depends on a third belief/attitude.
 - Polar $(-1, +1)$ spin state assumption; some beliefs may be better treated as 'on/off', where the 'off' state has *no* effect on other spins, rather than a negative effect.
 - More generally, effect sizes may vary depending on specific spin states.

Chapter 8

Climate beliefs dataset

In this chapter we outline the context and construction of the dataset used for model calibration in Chapter 4. We will first describe the broader context and complexities of the dataset underlying this study, before outlining our data validation and cleaning methods in Chapter 8.2. Finally, in Chapter 8.3 we detail the construction of the targeted dataset used for model calibration (Chapter 4).

While many empirical studies on belief systems rely on cross-sectional data (Lee et al., 2025; Powell et al., 2023; van Noord et al., 2025), longitudinal data is necessary to capture changes in individuals’ internal cognitive states over time. In fact, repeated per-individual observations are essential to our adopted method for model calibration. We are fortunate, in this study, to have had access to a longitudinal dataset on climate beliefs and attitudes in USA as our primary data resource.

The longitudinal study (**CITE**) comprises six waves of responses from individuals residing in the United States, collected between April 2020 and March 2023. The assessed dimensions include general demographic information (e.g., age, gender, education, and financial status), beliefs, attitudes, and experiences relating to concurrently-salient topics such as COVID-19, climate change, or the 2020 US presidential election, and support for hypothetical policies. In addition to a wide-form table of survey response data, the dataset includes a codebook which specifies, for each survey item, the question text, occurrence in survey waves, and conditional display logic where applicable. At present, the codebook is limited to Waves 1–5, i.e., excluding the sixth (final) wave.

However, the dataset is not without complexities. Survey participation varies, with each participant responding to a (possibly non-contiguous) subset of waves. Figure 8.1 shows the number of participants who have responded to each combination of survey waves, with a minimum count of 1500. While each individual wave has a relatively

high response rate (roughly 4000–5000), this rapidly drops off when other waves are considered. For instance, only ~2500 individuals responded to Waves 1 and 2, and only ~1900 of these individuals also responded to Wave 3.

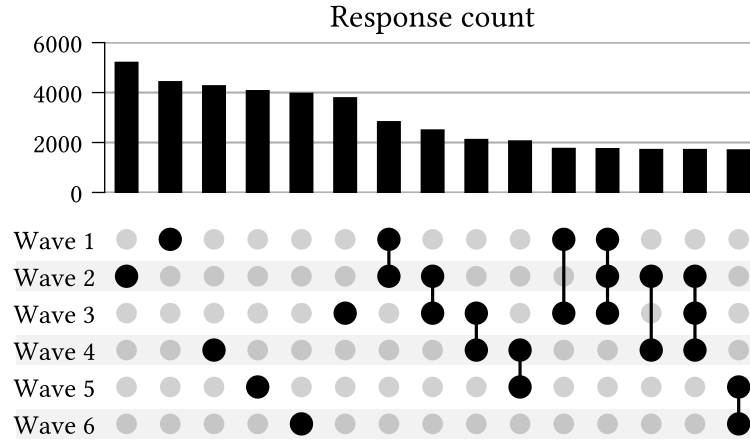


Figure 8.1: Number of repeat participants for different wave combinations. Combinations with total responses below 1500 have been removed.

Survey content also varies between waves and participants, both with regards to *which* questions are included, and *how* they are presented. Certain questions are displayed only in particular waves, or only to either repeating or new participants. Some questions are shown conditionally based on an individual’s responses to prior questions (within the same wave). Others vary according to survey treatment conditions, such that individuals in different treatment groups are presented different variants of a question. This survey logic is not always executed correctly; in some cases participants are shown survey questions incorrectly (e.g., ‘new’ participants shown questions intended only for ‘repeating’ participants). We discuss this issue further in Chapter 8.1.

Question format, text, and response schemas also occasionally change between survey waves. Changes in response schema are less common, however, and typically affect items with categorical responses. With a view to constructing the calibration dataset for our experiments, we note that most response schemas are not binary and therefore require binarisation (see Chapter 8.4).

Casting our attention to the survey timing, we examine the survey response dates for each wave (Figure 8.2) and the distribution of interval durations between consecutive-wave responses across individuals (Figure 8.3), i.e., the ‘inter-response time’.

In Figure 8.2 we observe that the survey waves occur with irregular spacing and duration. Some consecutive pairs of waves are much closer than others. For instance, notice that:

- (i) The duration from the start of Wave 1 until the end of Wave 3 is shorter than the duration *between* Waves 5 and 6, and
- (ii) The duration spanned by Wave 4 alone exceeds that of the interval spanned by both Waves 1 and 2.

This poses a potential problem for model calibration, since the non-equilibrium belief model (defined in Chapter 3) operates on the assumption that samples are equispaced.

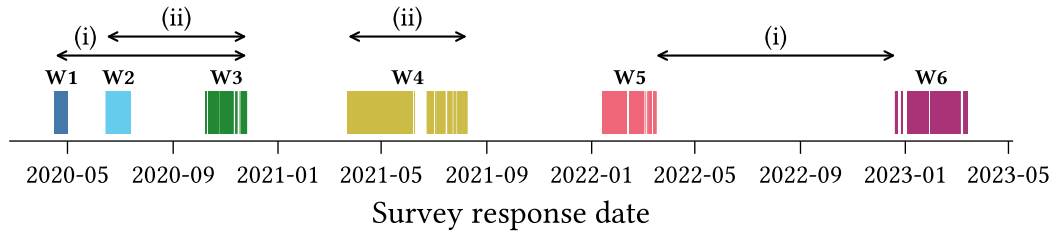


Figure 8.2: **TODO**

We see the irregular spacing reflected also in the inter-response time distributions. For each pair of consecutive waves, we observe substantial variation in the time between responses for different participants.

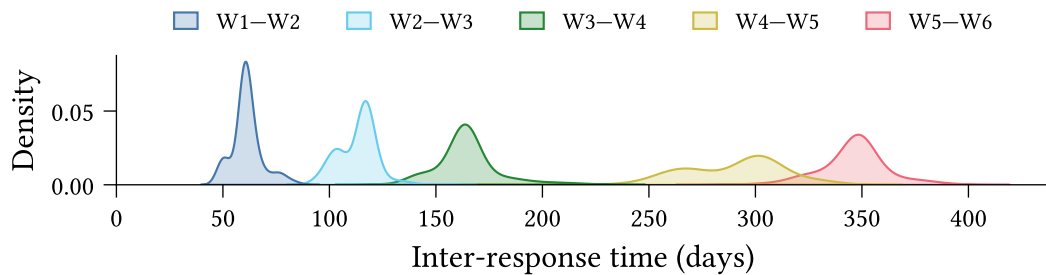


Figure 8.3: Between-response time distribution

Finally, we note that the time interval spanned by the longitudinal dataset includes several notable events which could reasonably be expected to influence — and confound — the dynamics of beliefs and attitudes in myriad contexts. These include the COVID-19 pandemic, which arrived in the US only three months prior to the first survey wave (Holshue et al., 2020), the 2020 US Presidential Election which occurred during Wave 3, the January 6 United States Capitol Attack, which occurred between Waves 3 and 4, ...

8.1 Data Validation

The complexity of the climate attitudes survey, as outlined above, necessitates a rigorous approach to data validation, both to identify errors in the expected schema and to ensure that the data matches our expectations.

We have implemented a general validation pipeline comprising three stages:

1. **Type-level validation:** The set of columns is exactly as expected, and all columns have the correct data type.
2. **Response-value validation:** All *non-null* responses are valid according to the schema specified in the codebook.
3. **Null-value validation:** Responses are null if, and only if, we expect them to be null.

8.1.1 Type-level validation

Type-level validation ensures that the *observed* data schema matches the *prescribed* schema. Since response data types vary between questions (see Table 8.1), the type-checking must be flexible and capable of handling complex data types.

For instance, categorical multiple-choice responses are represented using a `list[Enum]`, where `Enum` is a question-specific enum-type¹. Type validation for a multiple-choice question thus requires checking: (i) that the response column comprises `list`'s, and (ii) that all list elements belong to the set of values defined by the `Enum`.

Response schema	Raw type	Coerced type
Text	<code>str</code>	<code>str</code>
Single response (ordinal)	<code>int</code>	<code>int</code>
Single response (categorical)	<code>int</code>	<code>Enum</code>
Multiple response	<code>str</code>	<code>list[Enum]</code>
Numeric	<code>float</code>	<code>float int</code>

Table 8.1: **TODO**

¹An enum is a type defined by a finite set of allowable values. In our case the values are human-readable strings. For instance, the `dem_urban` survey question, which asks ‘What kind of area do you live in?’ has responses with data type described by the enum `{Urban, Suburban, Rural}`.

8.1.2 Response-value validation

Response-value validation then ensures that all non-null response values are valid according to the survey codebook. For most variables this is straightforward. Numeric and single-response ordinal variables typically have a defined range (e.g., $18 \leq \text{age} \leq 99$, or $1 \leq x \leq 5$ for a 5-point Likert scale variable). Text-entry responses are always considered valid.

Single-response categorical questions have `Enum` type, so type-level validation is sufficient to ensure that the responses do not contain any values outside the set defined by the `Enum`. However, the allowable values occasionally change between survey waves. Hence response-value validation is also required for categorical single-response and multiple-response variables, and in general must handle schema variation between waves.

8.1.3 Null-value validation

Finally, null-value validation ensures that responses are null if, and only if, they are expected to be null. For a question Q , the response of a participant P in wave W is permitted to be null, iff, at least one of the following four conditions is true:

- N1. Q is not asked in wave W ,
- N2. In wave W , Q is only shown to *new* participants, but P is a *repeating* participant (or vice versa),
- N3. In wave W , Q is only shown to a treatment group of which P is not a member, or
- N4. In wave W , Q is conditionally shown to participants based on their response to a distinct question Q' , and P 's response to Q' does not satisfy the condition.

Formally we can write the condition for P 's response R being null as:

$$\text{null}(R) \iff (\text{N1} \vee \text{N2} \vee \text{N3} \vee \text{N4}) \tag{8.1}$$

Null-value validation then consists in identifying the set of responses $\mathbf{R}$ such that for $R \in \mathbf{R}$, exactly one of $\text{null}(R)$ and $(\text{N1} \vee \text{N2} \vee \text{N3} \vee \text{N4})$ is true. Validation succeeds if $\mathbf{R}$ is empty. Otherwise the members of $\mathbf{R}$ are counterexamples to the above conditions.

8.1.4 Implementation

We implement type-level and simple response-value validation using the [Pandera](#) Python library for DataFrame validation (Bantilan et al., 2022). We implement manual validation checks for more complex response-value cases, such as questions whose response schema varies between waves.

The null-value validation logic, i.e., the process of testing for contradictions Equation 8.1, is also manually implemented. Conditions N1 and N2 are derived automatically from the survey codebook. Conditions N3 and N4 must be manually specified, as the codebook currently does not have a standardised method for describing conditional display logic.

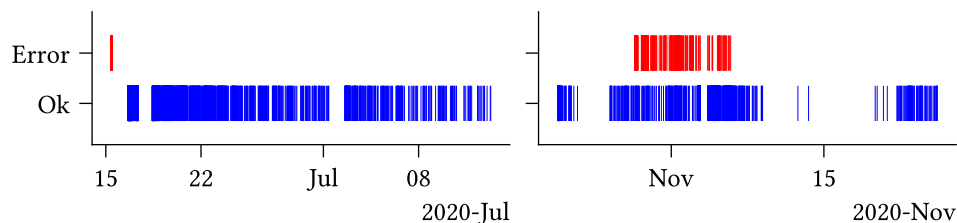
8.1.5 Validation results

At the time of writing, approximately 25% of the complete survey question set for Waves 1–5 has been validated, including all survey questions which were considered for the targeted calibration dataset (Chapter 8.3). Since Wave 6 is currently undocumented in the survey codebook we can reasonably validate neither the data schema, nor the presence of null-values. We have therefore not yet validated any of the data from Wave 6, and exclude this wave from the present study.

All type-level and response-value validation checks succeed, providing a strong guarantee that the data schema matches our expectations per the codebook. We do, however, encounter several problems during null-value validation.

In some cases, these were due to errors in the codebook itself. These errors are relatively straightforward to identify from the null-value validation results, since they often affect all individuals in a particular wave. For instance, if the codebook specified that a question is not presented in Wave 1, yet all responses are non-null, then this most likely indicates an error in the codebook. While some cases are more subtle, such as where a treatment class is misspecified, when arising due to a codebook error we still expect the contradictions to affect a well-defined subset of the population (in this case, the individuals in a particular treatment class).

In other cases we identify contradictions which are due to errors in the survey process, which caused certain survey questions to be displayed to some individuals in conditions which did not satisfy the specified survey logic. Some such cases were systematic, affecting all individuals until the survey process was updated; however, in other cases we have not been able to identify the source of the error.

Figure 8.4: **TODO**

We exclude all responses which fail the null-value validation from further analysis in the data cleaning stage (Chapter 8.2.2).

The described validation process is primarily concerned with ensuring that the data schema — comprising types and values — aligns with our expectations as per the survey codebook. While comprehensive in this regard, the process does not account for *all* possible forms of errors or inconsistencies. One category which is currently unaccounted for, but is worth mentioning, comprises inconsistencies in the responses from a given individual over multiple waves. For instance, the dataset includes several cases in which individuals presented the question:

“Were you born in the United States?”

respond with ‘Yes’ in one wave, but ‘No’ in another. Validating data for inconsistencies such as this requires careful consideration on a question-by-question basis to assess response types for conflicts. We consider this category beyond the scope of this study, describing it here only to illustrate the bounds of the above validation process.

8.2 Normalisation, cleaning, and transformations

We apply pre-processing steps to make the dataset amenable to downstream analysis, which are divided into three categories:

- **Normalisation:** Ensuring the data schema (variable names, types, and values) conform to consistent standards.
- **Cleaning:** Filtering erroneous or otherwise-problematic data.
- **Transformation:** Structural dataset enrichments, including constructed data variables and improved organisation of information.

8.2.1 Normalisation

Variable names in the original dataset are subject to some inconsistency, featuring a mixture of `snake_case`, `CamelCase`, `dot.case`, and `kebab-case`. We therefore first convert the names of all variables specified in the codebook to a standard format, making all names lowercase, and replacing hyphens and periods with underscores.

The dataset also contains several columns which are not described in the codebook, often containing survey metadata, the functions of which are typically indicated using a variable name prefix. For instance, the `Group_` prefix indicates that a column describes treatment group membership. We do not normalise these variable names here, so as to maintain identifiability for later transformations (Chapter 8.2.3).

Secondly, we coerce all variable data types to a standard set. In most cases this is straightforward. Dates and times represented as strings are coerced to timezone-free `datetime` types. Non-floating-point numeric values (which constitute most survey response types) are coerced to 64-bit integers.

Categorical variable responses are originally represented as integers, which presents three issues. Firstly, it imposes an ordinal structure which does not, in general, make sense for categorical variables. For instance, consider the first four response options to a survey question asking participants about their recent experience with different forms of extreme weather:

1. Severe winter storm,
2. Hurricane,
3. Tornado,
4. Heat wave

One may devise any number of reasonable ordinal interpretations over this set (e.g., ‘highest average risk of property damage’, or ‘annual emergency response cost’), yet such interpretations are context-dependent and not inherent to the categories themselves. Recording categorical responses using numeric types therefore risks spurious interpretations. Second, integer values provide no information about the categories to consumers of the dataset, requiring that they cross-reference manually with the codebook. Thirdly, in some cases the mapping from categories to integers for a given question varies between survey waves. We account for all three of these issues by coercing categorical variables to `Enum` types, which preserve information about the underlying categories, and do not assume an ordinal structure.

Multiple-selection survey items are more complex still. These are represented as strings comprising comma-separated integers, whose values typically refer to non-ordinal categories. We coerce these to `list[Enum]` types, which have all the benefits of the

categorical enums discussed above, and enable simpler programmatic analysis (e.g., testing for set membership, determining differences in response sets between waves).

Finally, we perform a series of value updates to ensure consistency between responses to different questions. This consists in replacing all empty string responses with null values², and re-mapping integer columns (both numeric and ordinal) such that the minimum value is zero.

8.2.2 Cleaning

Following schema normalisation, we then perform a cleaning stage, which primarily includes filtering out invalid or problematic data. We only filter out entire survey responses, rather than answers to specific survey items. Survey responses may be removed for any one of three reasons.

Firstly, we remove responses where the `participant_id` column is null, such that the individuals cannot be traced across survey waves. Secondly, we remove any responses from participants who fail the survey validity check in *any* of their participation waves. Thirdly, we remove any responses which fail either of the response-value or null-value validation checks described in the previous section (Chapter 8.1). We recognise that these issues are not necessarily problematic in every research context (e.g., cross-sectional studies are unconcerned with tracing an individual's responses across waves), and thus have made each condition optional in the dataset construction code.

8.2.3 Transformation

Finally, we apply several transformation steps with the goal of simplifying manipulation and analysis of the existing dataset, or enriching it with additional information.

The original dataset includes two survey items querying participants' alignment with the two largest US political parties (i.e., the Republicans and the Democrats). The first item asks whether the participant whether they consider themselves as *Republican*, *Democrat*, or *Independent*. Those who respond *Independent* are then presented a follow-up question asking whether they *lean* toward either (or neither) party. We replace these two survey items with a single variable combining the responses, with possible values:

Democrat $\prec$ Leaning Democrat $\prec$ Independent $\prec$ Leaning Republican $\prec$ Republican

²Note that this particular value mapping must happen *after* null-value validation (Chapter 8.1).

While this supplements the first item with additional information, we note two potential limitations of this transformation. Firstly, some participants who respond *Republican* or *Democrat* to the first question may more accurately describe themselves as only leaning in a given direction, if they had been aware of the (conditional) second question. Therefore the ‘Leaning’ responses are perhaps better interpreted as ‘Independent with Republican/Democrat tendencies’.

Secondly, our implicit assumption that *Independent* is a midpoint between *Republican* and *Democrat* would misrepresent individuals who consider themselves ‘right-of-Republican’ or ‘left-of-Democrat’. However, the available data does not allow us to distinguish such cases. Furthermore, this issue is also present in the original first question.

Our second transformation concerns survey questions with different variants, which are conditionally displayed based on treatment group membership. The original dataset includes, for each treatment group associated with a given question, a separate response column, as well as a binary indicator column describing group membership. Since group membership is always mutually exclusive in the dataset, we coalesce these various columns, replacing them with singular response and group membership columns. The group membership column contains integer indexes as opposed to the original binary indicators. In some cases, different treatment groups have an associated numerical value. For instance, the *ccSolveX* items query support policies compensating communities affected by climate change, with the treatment group specifying the cost (X) of such a policy to the participant. In these cases we supplement the dataset with an additional column containing the associated numerical value, simplifying downstream analyses.

Finally, we improve the dataset organisation by constructing a database comprising tables for survey item and question metadata and responses, as well as participants themselves (Figure 8.5). The requirement for such a database arose during the construction of the climate beliefs dataset used for model calibration (Chapter 8.3). The original data, comprising a codebook (a spreadsheet) and an RData file containing rows of participant responses, is not immediately amenable to our task, which requires reasoning about both survey logic and participation in tandem. The constructed database primarily serves to link survey metadata (extracted from the codebook) with survey response data.

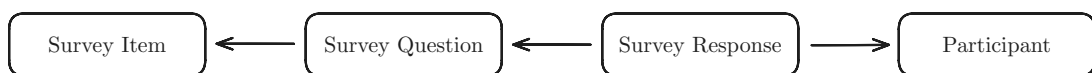


Figure 8.5: The survey database comprises tables for survey items (underlying concepts assessed in the survey), survey questions (the particular mode in which an item is assessed in a given wave or treatment condition), survey responses, and participants.

Recognising the hierarchical structure of the longitudinal dataset, in which a single survey item may vary in question text or response schema between waves or treatment conditions, we separate the concepts of *survey items* (the underlying concept being assessed) and *survey questions* (the particular mode of assessment). Survey questions are associated with particular waves, participant types (i.e., new or repeating), and treatment conditions, but may be related under a common survey item. *Survey responses* are associated with a particular question as well as a *survey participant*. We include a separate table for participants which records wave participation as well as the initial wave in which a participant has responded.

The database is stored on disk as a collection of parquet files (*Parquet*, n.d.), which preserve complex data type information and are supported in both R (e.g., using [Arrow](#)) and Python (e.g., using [Pandas](#) or [Polars](#)).

8.3 Climate beliefs dataset

TODO:

- We exclude sixth wave due to no codebook

In light of the complexities and breadth of content of the **placeholder dataset**, we construct a smaller, targeted dataset of beliefs and attitudes relating to climate change, which we expect – on theoretical grounds – to exhibit interdependent behaviour. We will refer to this targeted dataset as the **climate beliefs dataset**. The primary motivator for constructing this dataset is the calibration of the non-equilibrium belief system model (Chapter 4.3).

We aim for 7–10 variables in the final set, so as to keep both the number of model parameters and parameter uncertainty sufficiently small. We first filter the survey items to consider only those which assess cognitive states (e.g., beliefs, attitudes, opinions, stances). We further restrict our attention to items which either relate directly to climate change, or which are expected to influence climate-related cognitive states (e.g., political alignment).

Within this restricted set of survey items our intention is to select a collection of variables which: (i) exhibit related behaviour, (ii) maximise the number of observations, and (iii) satisfy the requirements of the parameter estimation method used for model calibration. We will discuss the latter two requirements now, and return to the first shortly. We require at least two waves of responses from each participant, as the model is defined by a time-dependent Markov process, and thus parameter estimation involves

fitting a conditional probability distribution where each belief state depends on the previous one. Moreover, each participant's responses should be equispaced, with spacing being consistent across individuals, and the dataset must contain no missing values. In addition to these requirements, we prioritise including several variables of interest:

1. **CC Worry:** Concern about current and future climate change
2. **CC Others Worry:** Belief regarding *others'* level of worry about climate change
3. **CC Human:** Belief that climate change is caused by human activities

The first is generally considered influential in shifting related attitudes, while the second is expected to be influential on the first, as a perceived descriptive norm. Finally, while most individuals in the USA believe in the *existence* of climate change, there is relatively greater variation in beliefs about its *causes*. Hence the third item is an interesting target for intervention studies. (**TODO: Add citations**)

Given the above requirements and the objective to maximise the number of observations, we identify Waves 3 & 4 as suitable candidates. These waves include 1693 repeat observations (excluding survey errors as described in the previous subsection). After removing items with no substantial correlations, and small groups of items which exhibit only internal correlations, we identify 17 relevant survey items, comprising both beliefs (epistemic positions, Table 8.2) and attitudes (qualitative evaluations, Table 8.3).

TODO: Perhaps include this as a subsection/callout, and make a paragraph for each question to specify the question text and schema. Then include a summary table with the item name, schema, coding.

Item	Question text	Response schema
CC Real	Do you think that climate change is happening?	• Yes • No • Don't know
CC Human (*)	Do you think rising temperatures are a result of human activities, natural causes, or both?	• Not happening • Natural causes • Human activities • Both
CC Impact (world)	How much do you think climate change is currently harming the world in general?	• Not at all • Only a little • A moderate amount • A great deal
CC Impact (wealthy)	How much do you think climate change is currently harming wealthy communities within the United States?	• Not at all • Only a little • A moderate amount • A great deal
CC Impact (poor)	How much do you think climate change is currently harming poor communities within the United States?	• Not at all • Only a little • A moderate amount • A great deal
CC Impact (comm)	How much do you think climate change is currently harming your local community?	• Not at all • Only a little • A moderate amount • A great deal
CC Others Worry	How worried do you think most Americans are about global warming/climate change these days?	• Not at all worried • Not very worried • Somewhat worried • Very worried

Table 8.2: * Re-mapped to: includes/does not include human activities

Item	Question text	Response schema
CC Worry	How worried are you about current and future global warming/climate change?	• Not at all worried • Not very worried • Somewhat worried • Very worried
Weather Worry	How worried are you about an extreme weather event or natural disaster happening to you personally in the next year?	• Not at all • Only a little • A moderate amount • A great deal
CC Responsibility	It is important that individuals take action on issues of climate change.	5-point Likert agreement scale
CC Scientists	Scientists with appropriate expertise should guide how we respond to climate change	5-point Likert agreement scale
Policy: ICA (*)	Do you favour an international agreement committing the USA and other countries to reduce their carbon emissions?	• Yes and legally binding) • Yes but <i>not</i> legally binding) • No
Policy: Tax fuel	How much do you support/oppose a tax on the production/distribution of carbon-based fuels?	5-point Likert support scale
Policy: Auto	How much do you support/oppose stronger carbon emissions standard for car manufacturers?	5-point Likert support scale
Policy: Env. Reg.	Which statement comes closer to your views?	• Stricter environmental regulations cost too many jobs and hurt the economy • Stricter environmental regulations are worth the cost
Political affiliation (**)	In politics today, do you consider yourself a:	• Republican • Leaning Republican • Independent • Leaning Democrat • Democrat
Political ideology	What is your political ideology?	• Very conservative • Conservative • Moderate • Liberal • Very liberal

Table 8.3: * Re-mapped to: Yes/No

** Constructed from separate items for identification and leaning

To reduce the set of candidate variables to the target range of 7–10, we proceed to identify groups of similar or redundant variables which may be removed or combined

into interpretable index variables. We use two primary measures to investigate similarity: pairwise partial correlation, and vector autoregression (Epskamp, van Borkulo, et al., 2018; Epskamp, Waldorp, et al., 2018).

Partial correlation measures the correlation between a pair of variables X and Y after conditioning on all remaining variables Z :

$$\rho(X, Y) = \text{Corr}(X, Y \mid Z) \quad (8.2)$$

In particular, $|\rho(X, Y)|$ is non-zero if X and Y exhibit *linear* correlation which is not shared with the remaining variables in Z . For instance, consider the fork structure $X \leftarrow A \rightarrow Y$ where X and Y are probabilistic linear functions of A . If the value of A is unknown, then X and Y are dependent. However, once A is known, the dependency structure is broken. Thus $\text{Corr}(X, Y)$ is nonzero, while $\rho(X, Y) = 0$.

Figure 8.6 displays the pairwise partial correlation between each pair of variables in the filtered set. Note that we have coded all variables such that the partial correlation matrix entries are predominantly positive. Cells with magnitude below 0.05 have been masked for visual clarity. Rows and columns are ordered using hierarchical clustering.

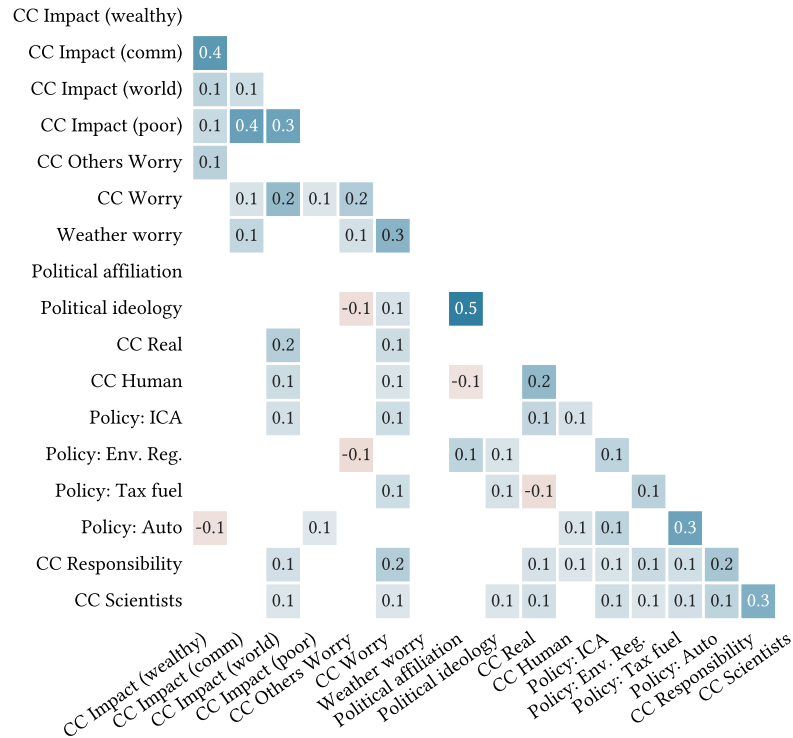


Figure 8.6: Pairwise partial correlation for the filtered candidate set (Table 8.2, Table 8.3). Cells with $|\rho(X, Y)| < 0.05$ are masked for visual clarity. Rows and columns are ordered using hierarchical clustering over the distance matrix: $d_\rho(X, Y) = 1 - |\rho(X, Y)|$.

Partial correlations are limited, however, in that they capture only instantaneous (or ‘contemporaneous’) relations between variables, while the non-equilibrium belief system model is concerned with temporal relationships. Therefore we also examine the temporal and contemporaneous networks obtained using vector autoregression (VAR) with a $\delta t = 1$ time lag (Brandt & Williams, 2007). These networks are derived from the solution to the following regression problem, described in Epskamp, Waldorp, et al. (2018):

$$\begin{aligned} \mathbf{y}^t &= \mathbf{B}\mathbf{y}^{t-1} + \varepsilon^t \\ \varepsilon^t &\sim \mathcal{N}(\mathbf{0}, \Theta) \end{aligned} \quad (8.3)$$

Where $\mathbf{y}^t$ is the vector of observed variable values for each individual at time t . The *temporal network* matrix $\mathbf{B}$ comprises linear next-state predictors, while the *contemporaneous network* matrix $\mathbf{K} = \Theta^{-1}$ describes the pairwise conditional correlations which remain after accounting for temporal effects. The entries of $\mathbf{K}$ are analogous to partial correlations, constituting the conditional correlation between X and Y given the remaining variables.

Figure 8.7 shows the contemporaneous and temporal network matrices for the filtered set of candidate variables. The contemporaneous matrix is symmetric, thus the upper triangle is not shown.

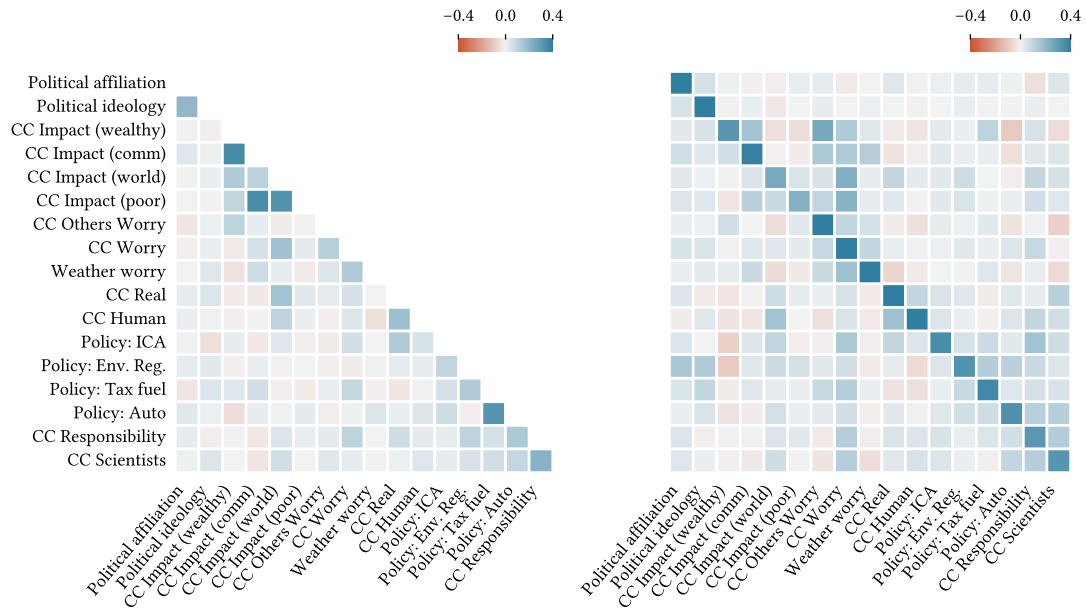


Figure 8.7: Contemporaneous (left) and temporal (right) network matrices for the filtered candidate set (Table 8.2 Table 8.3). Entries in row i of the temporal matrix are the regression predictors for the subsequent value of variable X_i . Rows and columns are ordered using hierarchical clustering on the distance matrix: $d_{\mathbf{K}}(X, Y) = 1 - |\mathbf{K}|$.

We observe some apparent commonalities between the partial correlation and VAR networks. First, the *CC Impact* variables exhibit relatively strong internal relations. External relations with variables outside this set are typically weaker, with the exception of *CC Impact (world)*, which has several non-trivial partial correlations with other variables, including other climate-related beliefs. All variables in this group, except *CC Impact (wealthy)*, also have non-trivial partial correlations with *CC Worry* — one of our variables-of-interest. Second, the policy variables, as well as *CC Responsibility* and *CC Scientists*, also exhibit substantial connectivity in the partial correlation network. This is diminished in the contemporaneous VAR network, but visible in the clustering of the temporal network. The similarities among these variables in the rows and columns of the temporal matrix indicate that they fill similar roles as predictors, and are also predicted similarly. Thirdly, *Political affiliation* and *Political ideology* are highly related in the partial correlation matrix, and also exhibit similar behaviour in the temporal network.

Based on these observations, we construct three index variables: (i) *CC Impact*, comprising the four climate impact beliefs, (ii) *Politics*, comprising the measures of political affiliation and ideology, and (iii) *CC Action*, comprising the four policy variables, as well as *CC Responsibility* and *CC Scientists*.

8.4 Binarisation

Prior to model fitting, we binarise all variables, mapping data values to the domain $\{-1, +1\}$. Before binarising, we shift each variable such that the *survey midpoint* (e.g., ‘3’ on a 5-point Likert scale) aligns with 0, and rescale each variable such that the minimum and maximum possible values map to -1 and $+1$ respectively. For variables

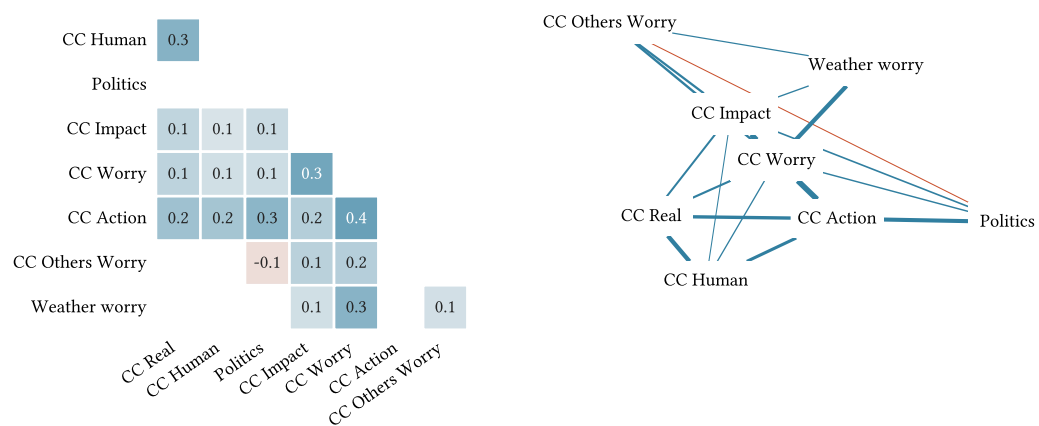
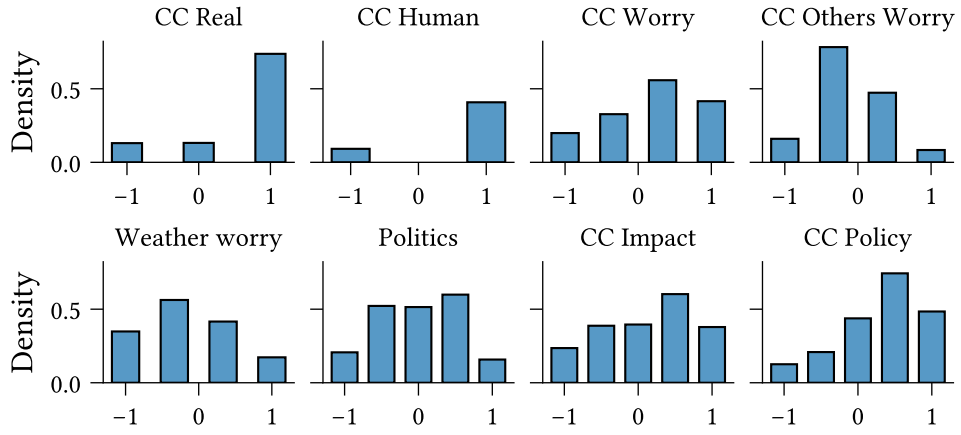


Figure 8.8: TODO

Figure 8.9: **TODO**

with no well-defined midpoint, such as those with an even number of possible responses, we shift such that the minimal and maximal values are at equal distance from 0.

We use a smooth binarisation process to robustly handle data points which are zero, or close to zero. This involves perturbing each data value $x \in \mathbb{R}$ by an independent noise term $\varepsilon \sim \mathcal{N}(0, \sigma)$ before thresholding. Figure 8.10 illustrates this process for a negative value of x and given choice of σ .

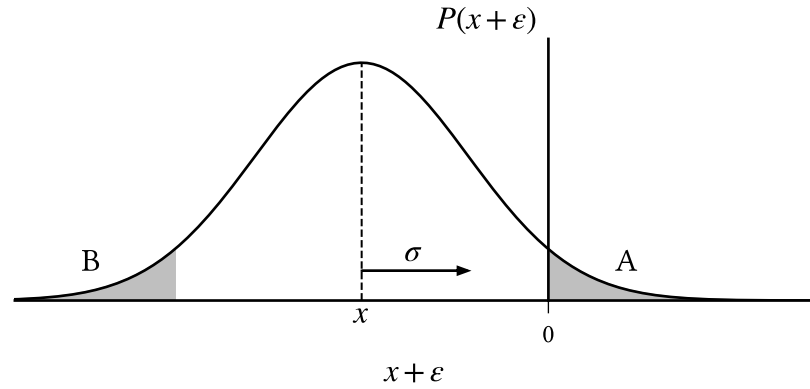


Figure 8.10: $x \in \mathbb{R}$ is smoothly binarised to $\{-1, +1\}$ by thresholding $x' = (x + \varepsilon)$ where $(x + \varepsilon) \sim \mathcal{N}(x, \sigma)$. Negative values are mapped to $+1$ with probability $P(x' > 0) = A = B = P(\varepsilon < x)$, which increases with σ and $|x|^{-1}$.

Observe that x is mapped to $+1$ if, and only if, ε is sufficiently large, such that $x + \varepsilon > 0$ (region A), or equivalently when $\varepsilon < x$ (region B). The probability that x is mapped to $+1$ is then

$$P(x \mapsto +1) = \Phi\left(\frac{x}{\sigma}\right) \quad (8.4)$$

where Φ is the standardised normal cumulative distribution function. This probability is small when x has large magnitude, or when σ is small. For the purposes of our experiments, we choose $\sigma = 0.203$ such that a ‘weakly oppose’ response to a 7-point Likert scale³ (value 1/3) is mapped to +1 with probability 0.05.

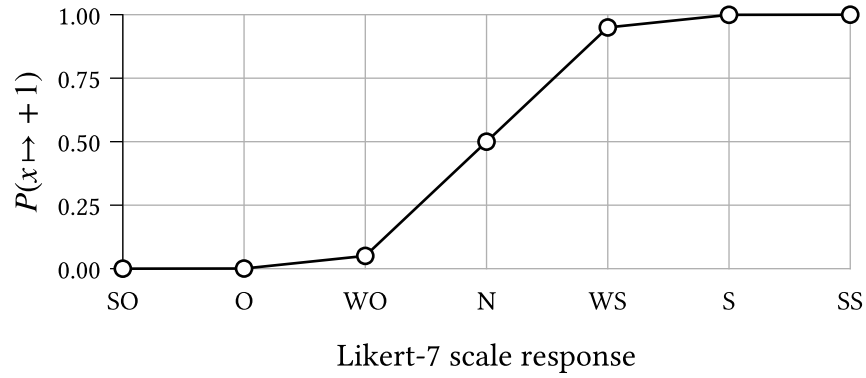


Figure 8.11: The probability of binarisation to +1 for each possible response to a Likert-7 scale survey question, given $\sigma \approx 0.2$. Ordered responses are: *strongly oppose* (SO), *oppose* (O), *weakly oppose* (WO), *neutral* (N), *weakly support* (WS), *support* (S), *strongly support* (SS).

TODO: Consider specifying Likert-7 scale options in a table, and referencing that.

³The oppose/support 7-point Likert scale has possible responses: strongly oppose, oppose, weakly oppose, neutral, weakly support, support, strongly support.

Chapter 9

Related work

9.1 Plan

1. Psychological/philosophical theories of beliefs and attitudes
 - Cognitive consistency:
 - Heider (balance theory) (Heider, 1946), Insko & Schopler (triadic consistency) (Insko & Schopler, 1967)
 - Festinger (cognitive dissonance) (Festinger, 1962)
 - Fishbein (Fishbein, 1966)
 - Quine (Quine, Willard van Orman & Ullian, Joseph Silbert, 1978)
2. Belief system modelling:
 - Schools of thought:
 - Triadic consistency (Rodriguez et al., 2016)
 - Parallel constraint satisfaction (Monroe & Read, 2008)
 - Bayesian networks (Cook & Lewandowsky, 2016; Powell et al., 2023)
 - Structural equation modelling?
 - Regularised partial correlation networks
 - Ising
 - CAN (Dalege et al., 2016)
 - Social influence: Network of belief (Dalege et al., 2025), hierarchical Ising model (van der Maas et al., 2020)
 - What theory/assumptions does each model draw on?
 - Difference between Latent variable and empirical network models (Dalege et al., 2016)

-
- Case studies — how have the models been applied?
 - Theoretical explanations for known phenomenon
 - Analysis of survey data
3. Interventions
- Modelling, testing effects (Powell et al., 2023)
 - Theoretical (Dalege et al., 2025)
 - In-practice (The Workshop, 2024)
4. Inference
- Between-individual limitations (Brandt & Morgan, 2022)

Chapter 10

Conclusions and future work

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doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

Chapter 11

Ethics and Data Management

A new requirement for the thesis is that there must be a short section in which you reflect on the ethical aspects of your project. This requirement is related to one of the final objectives that a graduated student of the Master of Computational Science must meet: “The graduate of the program has insight into the social significance of Computational Science and the responsibilities of experts in this field within science and in society“. You don’t need to devote an entire chapter to this; a short section or paragraph is sufficient.

I acknowledge that the thesis adheres to the ethical code (<https://student.uva.nl/en/topics/ethics-in-research>) and research data management policies (<https://rdm.uva.nl/en>) of UvA and IvI.

The following table lists the data used in this thesis (including source codes). I confirm that the list is complete and the listed data are sufficient to reproduce the results of the thesis. If a prohibitive non-disclosure agreement is in effect at the time of submission “NDA” is written under “Availability” and “License” for the concerned data items.

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Appendix A

Parameter estimation

Log-likelihood

Equation 4.5

Expected log-likelihood

Equation 4.6

Partial derivatives of expected log-likelihood

Dataset

Binarisation probability

Equation 8.4